

VR3511F Data Sheet

VRender™ 3D

Data Sheet

MDVR3511FDS

Ver 1.00 / February 6, 2006

VR3511F

*Data Sheet for VR3511F 3D Graphic Application
Processor*

Copyright © 2001/2/3/4/5, MagicEyes Digitals Co.

MAGIC EYES

All rights reserved. No part of this document may be reproduced in any form without written permission from MagicEyes Digital Co. MagicEyes Digital Co. reserved the rights to change in its products or product specification to improve functions or design at any time without notice.

NOTICE

MAGICEYES INFORMS THAT THIS CODE AND INFORMATION IS PROVIDED "AS IS" BASE AND WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND/OR FITNESS FOR A PARTICULAR PURPOSE.

Important Notice

Information in this document is provided solely to enable system and software implementers to use MagicEyes products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

MagicEyes reserves the right to make changes without further notice to any products herein.

MagicEyes makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MagicEyes assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters which may be provided in MagicEyes data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. MagicEyes does not convey any license under its patent rights nor the rights of others. MagicEyes products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the MagicEyes product could create a situation where personal injury or death may occur. Should Buyer purchase or use MagicEyes products for any such unintended or unauthorized application, Buyer shall indemnify and hold MagicEyes and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that MagicEyes was negligent regarding the design or manufacture of the part.

© MagicEyes Digital

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electric or mechanical, by photocopying, recording, or otherwise, without the prior written consent of MagicEyes Digital.

HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu Seongnam-City, Gyeonggi-Do, Korea

TEL: 82-31-788-0300

FAX: 82-31-707-5765

HOME PAGE:

<http://www.mesdigital.com>

INDEX

1. Overview	5
2. PAD Description	7
3. Power Management Controller	19
4. Clock Controller	20
5. GPIO Controller	21
6. Memory Controller	22
7. Interrupt Controller	24
8. Timer	25
9. DMA	26
10. USB HOST CONTROLLER (UHC)	27
11. USB DEVICE CONTROLLER (UDC)	28
12. UART	30
13. PWM	32
14. I2C	33
15. ADC & TPC	34
16. SD/MMC Controller	35
17. NGPIO Controller	36
18. SSP/SPI	37
19. Wavetable Synthesizer	38
20. Audio Controller	39
21. VLD/VLC	40
22. MPEG4 Picture Decoder	42
23. 2D Graphic Controller	43
24. 3D Graphic Engine	44
25. Image Signal Processor (ISP)	45
26. Color Space Converter(CSC)	46
27. Multi Layer Controller (MLC)	47
28. Display Controller	50
29. ELECTRICAL DATA	51

1. Overview

VRender3D 32bit RISC microprocessor is designed to provide a cost-effective, low power, small die size and high performance micro-controller for hand-held devices and game applications. To reduce total system cost, VRender3D also provides the following : separate 16KB instruction and 16KB Data Cache, MMU to handle virtual memory management, Digital LCD Interface, 2-channel UART with handshake, 12-channel DMA, System Manager, 4-channel Timers, 2-channel PWM, 10bit ADC for touch-screen interface, IIC Controller, AC97/I2S codec Interface, USB device, Multimedia card interface,.

The VRender3D was developed using an ARM920T core, 0.18um CMOS standard cells and a memory compiler. Its low-power, simple, elegant and fully static design is particularly suitable for cost-sensitive and power sensitive applications.

An outstanding features of the VRender3D is 3D graphic accelerator, GTE, Wavetable Synthesizer.

By Providing complete set of common system peripherals, the VRender3D minimizes overall system costs and eliminates the need to configure additional components. The integrated on-chip functions that are described in this document include;

CPU

- ARM920T@200Mhz
- Data Cache : 16KByte
- Instruction Cache : 16KByte

MEMORY

- DDR-SDRAM
- x16, x32bit Interface
- 64/128/256/512Mbit SDRAM
- Static Memory
- x8, x16, Interface
- PCMCIA/CF/IDE Controller

Enhanced 3D Features

GTE (Geometric Transform Engine)

- Vector Processing Unit (24bit Floating point: s15e8)
- Embedded Projection, View frustum Clipping, Viewport scaling
- Programmable Feature
- add/mul/dp3/dp4/dp3clamp/dp4clamp/rsqrt/min/max...

TSE (Triangle Setup Engine)

- 24bit Floating Point Engine
- Various Primitive
- Triangle strip, Triangle fan, Triangle list, Rectangle list

Rasterizer

- Sub-Pixel Correction
- Max resolution: 1024x768
- Texture Mapping
- 2-stage multi-texturing
- Perspective Correction
- Sub Texel Accuracy
- Bilinear Filtering
- Internal Texture Cache : 8KByte
- Enhanced Prefetch Texture Algorithm
- MIPMAP
- Use MPEG Video Texture
- Z-Buffer, Alpha Blending, Dithering
- W-based pixel fog
- Render to texture

MPEG Video

MPEG4 Video Decoder

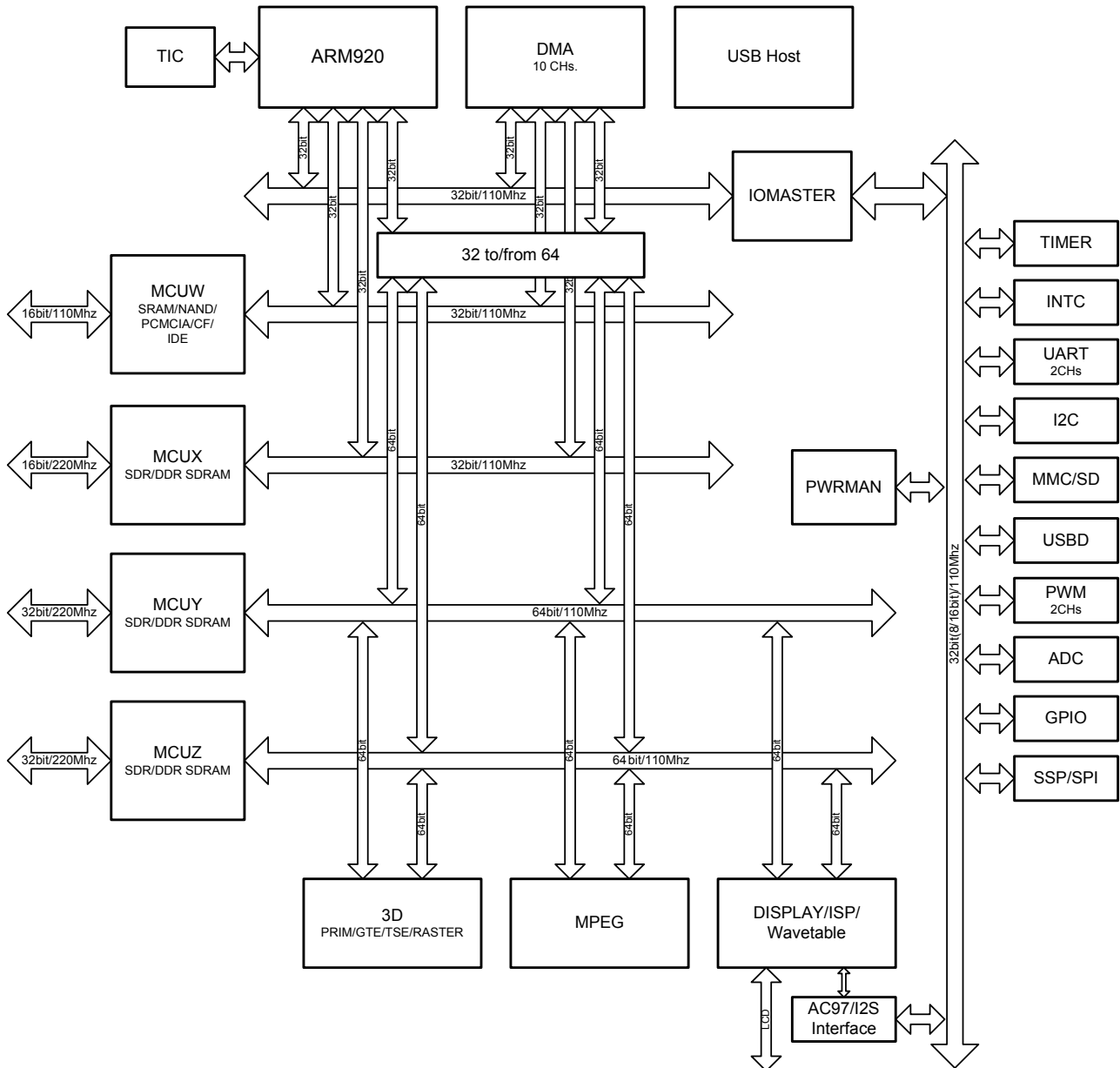
- Simple Profile, Level2 Decoder
- Up to 1024 x 1024 Resolution
- Supports MPEG Video Texture for 2D/3D
- 720x480 @30Hz real time decoding

Wavetable Synthesizer

- Output Format
- 16bit Stereo, ~44.1Khz
- Interface : AC97 Codec, I2S
- Up to 64 output channels
- Input Format
- 8/16bit PCM, 8bit u-law
- Effects
- individual volume/panning control
- individual 6 stage envelope control

Peripheral

- 10 CHs DMA
- 4 CHs Timer (Watchdog Timer)
- 2 CHs PWM (Pulse Width Modulation)
- Interrupt Controller
- USB 1.2 Host/Slave, I2C
- 2 CHs UART
- MMC / SD / NAND Flash
- AC97, I2S Codec Interface
- Camera interface (ISP)



2. PAD Description

2.1 PAD List

BALL	PIN NAME	I/O	Description	Function #1	Function #2	Reset	Type
C9	TCK	I	TCK (TAP Controller Clock) provides the clock input for the JTAG logic. A 10K pull-up resistor must be connected to TCK pin.	TCK		F1	PI
B9	TMS	I	TMS (TAP Controller Mode Select) controls sequence of the TAP controller's states. A 10K pull-up resistor has to be connected to TMS pin.	TMS		F1	PI
D8	TDI	I	TDI (TAP Controller Data Input) is the serial input for test instructions and data. A 10K pull-up resistor has to be connected to TDI pin.	TDI		F1	PI
A7	TDO	O	TDO (TAP Controller Data Output) is the serial output for test instructions and data	TDO		F1	P8
E10	nTRST	I	nTRST (TAP Controller Reset) resets the TAP controller at start. If debugger is used. A 10K pull-up resistor has to be connected. If debugger is not used, nTRST pin must be issued by a low active pulse (Typically connected to nRESET)	nTRST		F1	PI
A8	TESTEN	I	Test Mode Enable 0: Normal Operation, 1: Test Mode	TESTEN		F1	PI
D9	TESTSCAN	I	For test purpose only 0: Normal Operation	TESTSCAN		F1	PI
C1	XTALIN	AI	Crystal Input for internal PLL. 10.000Mhz	XTALIN		F1	M26
B1	XTALOUT	AO	Crystal Output for Internal PLL	XTALOUT		F1	M26
D1	EXTCLK	I	External Clock Source for test purpose only. It has to be low (GND)	EXTCLK		F1	PI
A9	PWRXTALIN	AI	Crystal Input for Power Management Recommend : 32.768Khz	PWRXTALIN		F1	GP
A10	PWRXTALOUT	AO	Crystal Output for Power Management	PWRXTALOUT		F1	GP
E9	nBATF	I	Battery Fault, Probe for battery state (Does not wake up at power-off mode in case of low battery state). If it isn't used, it has to be High(3.3v). In VR3Dr2, Because It isn't supported, it must be connected to High.	nBATF		F1	PI
B8	nVDDF	I	VDD Fault, Probe for VDD state (Does not wake up at power-off mode in case of unstable power state). If it isn't used, it has to be High(3.3v) Because The Power-off mode (SLEEP, DEEP-SLEEP Mode) is not supported in VR3Dr2, This pin must be connected to High.	nVDDF		F1	PI
D7	nPORST	I	nPORST places VR3D into a known reset state. For a reset, nPORST must be hold to Low level for at least 4 PWRXTAL after power has been stabilized.	nPORST		F1	PI
E8	nRSTOUT	O	For external device reset control, It is activated by nPORST, Watchdog Reset, SW Reset.	nRSTOUT		F1	P8
C8	PWRON	O	1.8V core power on-off control signal. If Power-off mode is not used, PWRON pad is "don't care"	PWRON		F1	P8
E14	XA0/GPIOA0	IO	X-BUS DDR-DRAM Address : XA[12:0] (DDR DRAM Address bus for X-Bank) outputs the memory address of the DDR for x-bank / GPIO	XA0		GI	SB
D14	XA1/GPIOA1	IO	X-BUS DDR-DRAM Address / GPIO	XA1		GI	SB
C14	XA2/GPIOA2	IO	X-BUS DDR-DRAM Address / GPIO	XA2		GI	SB
C13	XA3/GPIOA3	IO	X-BUS DDR-DRAM Address / GPIO	XA3		GI	SB
E13	XA4/GPIOA4	IO	X-BUS DDR-DRAM Address / GPIO	XA4		GI	SB
D13	XA5/GPIOA5	IO	X-BUS DDR-DRAM Address / GPIO	XA5		GI	SB
F13	XA6/GPIOA6	IO	X-BUS DDR-DRAM Address / GPIO	XA6		GI	SB
C12	XA7/GPIOA7	IO	X-BUS DDR-DRAM Address / GPIO	XA7		GI	SB
E12	XA8/GPIOA8	IO	X-BUS DDR-DRAM Address / GPIO	XA8		GI	SB

A12	XA9/GPIOA9	IO	X-BUS DDR-DRAM Address / GPIO	XA9		GI	SB
A17	XA10/GPIOA10	IO	X-BUS DDR-DRAM Address / GPIO	XA10		GI	SB
F12	XA11/GPIOA11	IO	X-BUS DDR-DRAM Address / GPIO	XA11		GI	SB
D12	XA12/GPIOA12	IO	X-BUS DDR-DRAM Address / GPIO	XA12		GI	SB
B17	XBA0/GPIOA13	IO	X-BUS DDR-DRAM Bank Address : XBA[1:0] (DDR DRAM Bank Address for X-Bank) output bank address for DDR-DRAM / GPIO	XBA0		GI	SB
F14	XBA1/GPIOA14	IO	X-BUS DDR-DRAM Bank Address / GPIO	XBA1		GI	SB
C10	XCK	O	X-BUS DDR-DRAM Clock (DDR DRAM Clock for X-Bank)	XCK		F1	SD
B10	XnCK	O	X-BUS DDR-DRAM Inverting Clock (DDR DRAM Inverted Clock for X-Bank)	XnCK		F1	SD
C11	XCCKE	O	X-BUS DDR-DRAM Clock Enable	XCCKE		F1	SO
D15	XnCS/GPIOB0	IO	X-BUS DDR-DRAM Chip Select / GPIO	XnCS0		GI	SB
D16	XnRAS/GPIOB1	IO	X-BUS DDR-DRAM RAS (Row address strobe) / GPIO	XnRAS		GI	SB
E15	XnCAS/GPIOB2	IO	X-BUS DDR-DRAM CAS (Column address strobe) / GPIO	XnCAS		GI	SB
C16	XnWE/GPIOB3	IO	X-BUS DDR-DRAM Write Enable / GPIO	XnWE		GI	SB
E16	XLDM/GPIOB4	IO	X-BUS DDR-DRAM Data Mask : Lower Byte Data mask / GPIO	XLDM		GI	SB
D11	XUDM/GPIOB5	IO	X-BUS DDR-DRAM Data Mask : Higher Byte Data mask / GPIO	XUDM		GI	SB
C17	XLDQS	IO	X-BUS DDR-DRAM Data Strobe : Lower Byte Data strobe	XLDQS		F1	SB
E11	XUDQS	IO	X-BUS DDR-DRAM Data Strobe : High Byte Data strobe	XUDQS		F1	SB
A21	XD0	IO	X-BUS DDR-DRAM Data : XD[15:0], Data bus	XD0		Hi-Z	SB
B20	XD1	IO	X-BUS DDR-DRAM Data	XD1		Hi-Z	SB
A20	XD2	IO	X-BUS DDR-DRAM Data	XD2		Hi-Z	SB
B19	XD3	IO	X-BUS DDR-DRAM Data	XD3		Hi-Z	SB
A19	XD4	IO	X-BUS DDR-DRAM Data	XD4		Hi-Z	SB
B18	XD5	IO	X-BUS DDR-DRAM Data	XD5		Hi-Z	SB
A18	XD6	IO	X-BUS DDR-DRAM Data	XD6		Hi-Z	SB
B16	XD7	IO	X-BUS DDR-DRAM Data	XD7		Hi-Z	SB
A16	XD8	IO	X-BUS DDR-DRAM Data	XD8		Hi-Z	SB
B15	XD9	IO	X-BUS DDR-DRAM Data	XD9		Hi-Z	SB
A15	XD10	IO	X-BUS DDR-DRAM Data	XD10		Hi-Z	SB
B14	XD11	IO	X-BUS DDR-DRAM Data	XD11		Hi-Z	SB
A14	XD12	IO	X-BUS DDR-DRAM Data	XD12		Hi-Z	SB
B13	XD13	IO	X-BUS DDR-DRAM Data	XD13		Hi-Z	SB
A13	XD14	IO	X-BUS DDR-DRAM Data	XD14		Hi-Z	SB
B11	XD15	IO	X-BUS DDR-DRAM Data	XD15		Hi-Z	SB
P18	YA0/GPIOC0	IO	Y-BUS DDR-DRAM Address : YA[12:0] (DDR DRAM Address bus for Y-Bank) outputs the memory address of the DDR for Y-bank / GPIO	YA0		GI	SB
U22	YA1/GPIOC1	IO	Y-BUS DDR-DRAM Address / GPIO	YA1		GI	SB
T20	YA2/GPIOC2	IO	Y-BUS DDR-DRAM Address / GPIO	YA2		GI	SB
T19	YA3/GPIOC3	IO	Y-BUS DDR-DRAM Address / GPIO	YA3		GI	SB
F20	YA4/GPIOC4	IO	Y-BUS DDR-DRAM Address / GPIO	YA4		GI	SB
H19	YA5/GPIOC5	IO	Y-BUS DDR-DRAM Address / GPIO	YA5		GI	SB
E22	YA6/GPIOC6	IO	Y-BUS DDR-DRAM Address / GPIO	YA6		GI	SB
J20	YA7/GPIOC7	IO	Y-BUS DDR-DRAM Address / GPIO	YA7		GI	SB
H20	YA8/GPIOC8	IO	Y-BUS DDR-DRAM Address / GPIO	YA8		GI	SB
J19	YA9/GPIOC9	IO	Y-BUS DDR-DRAM Address / GPIO	YA9		GI	SB
R20	YA10/GPIOC10	IO	Y-BUS DDR-DRAM Address / GPIO	YA10		GI	SB

K17	YA11/GPIOC11	IO	Y-BUS DDR-DRAM Address / GPIO	YA11		GI	SB
K18	YA12/GPIOC12	IO	Y-BUS DDR-DRAM Address / GPIO	YA12		GI	SB
N18	YBA0/GPIOC13	IO	Y-BUS DDR-DRAM Bank Address : YBA[1:0] (DDR DRAM Bank Address for Y-Bank) output bank address for DDR-DRAM / GPIO	YBA0		GI	SB
R19	YBA1/GPIOC14	IO	Y-BUS Bank Address / GPIO	YBA1		GI	SB
K20	YCK	O	Y-BUS DDR-DRAM Clock (DDR DRAM Clock for Y-Bank)	YCK		F1	SD
L19	YnCK	O	Y-BUS DDR-DRAM Inverting Clock (DDR DRAM Inverted Clock for Y-Bank)	YnCK		F1	SD
L17	YCKE	O	Y-BUS DDR-DRAM Clock Enable	YCKE		F1	SO
P20	YnCS/GPIOB6	IO	Y-BUS DDR-DRAM Chip Select / GPIO	YnCS		GI	SB
M18	YnRAS/GPIOB7	IO	Y-BUS DDR-DRAM RAS (Row address strobe) / GPIO	YnRAS		GI	SB
N20	YnCAS/GPIOB8	IO	Y-BUS DDR-DRAM CAS (Column address strobe) / GPIO	YnCAS		GI	SB
M19	YnWE/GPIOB9	IO	Y-BUS DDR-DRAM Write Enable / GPIO	YnWE		GI	SB
U20	YLDM0/GPIOB10	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YLDM0		GI	SB
M17	YUDM0/GPIOB11	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YUDM0		GI	SB
N19	YLDM1/GPIOB12	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YLDM1		GI	SB
G20	YUDM1/GPIOB13	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YUDM1		GI	SB
U21	YLDQS0	IO	Y-BUS DDR-DRAM Data Strobe	YLDQS0		F1	SB
L18	YUDQS0	IO	Y-BUS DDR-DRAM Data Strobe	YUDQS0		F1	SB
P19	YLDQS1	IO	Y-BUS DDR-DRAM Data Strobe	YLDQS1		F1	SB
E21	YUDQS1	IO	Y-BUS DDR-DRAM Data Strobe	YUDQS1		F1	SB
Y22	YD0	IO	Y-BUS DDR-DRAM Data	YD0		Hi-Z	SB
W21	YD1	IO	Y-BUS DDR-DRAM Data	YD1		Hi-Z	SB
W22	YD2	IO	Y-BUS DDR-DRAM Data	YD2		Hi-Z	SB
V21	YD3	IO	Y-BUS DDR-DRAM Data	YD3		Hi-Z	SB
V22	YD4	IO	Y-BUS DDR-DRAM Data	YD4		Hi-Z	SB
T21	YD5	IO	Y-BUS DDR-DRAM Data	YD5		Hi-Z	SB
T22	YD6	IO	Y-BUS DDR-DRAM Data	YD6		Hi-Z	SB
R21	YD7	IO	Y-BUS DDR-DRAM Data	YD7		Hi-Z	SB
R22	YD8	IO	Y-BUS DDR-DRAM Data	YD8		Hi-Z	SB
P21	YD9	IO	Y-BUS DDR-DRAM Data	YD9		Hi-Z	SB
P22	YD10	IO	Y-BUS DDR-DRAM Data	YD10		Hi-Z	SB
N21	YD11	IO	Y-BUS DDR-DRAM Data	YD11		Hi-Z	SB
N22	YD12	IO	Y-BUS DDR-DRAM Data	YD12		Hi-Z	SB
M21	YD13	IO	Y-BUS DDR-DRAM Data	YD13		Hi-Z	SB
M22	YD14	IO	Y-BUS DDR-DRAM Data	YD14		Hi-Z	SB
L21	YD15	IO	Y-BUS DDR-DRAM Data	YD15		Hi-Z	SB
L22	YD16	IO	Y-BUS DDR-DRAM Data	YD16		Hi-Z	SB
K21	YD17	IO	Y-BUS DDR-DRAM Data	YD17		Hi-Z	SB
K22	YD18	IO	Y-BUS DDR-DRAM Data	YD18		Hi-Z	SB
J21	YD19	IO	Y-BUS DDR-DRAM Data	YD19		Hi-Z	SB
J22	YD20	IO	Y-BUS DDR-DRAM Data	YD20		Hi-Z	SB
H21	YD21	IO	Y-BUS DDR-DRAM Data	YD21		Hi-Z	SB
H22	YD22	IO	Y-BUS DDR-DRAM Data	YD22		Hi-Z	SB
G21	YD23	IO	Y-BUS DDR-DRAM Data	YD23		Hi-Z	SB
G22	YD24	IO	Y-BUS DDR-DRAM Data	YD24		Hi-Z	SB

F22	YD25	IO	Y-BUS DDR-DRAM Data	YD25		Hi-Z	SB
F21	YD26	IO	Y-BUS DDR-DRAM Data	YD26		Hi-Z	SB
D22	YD27	IO	Y-BUS DDR-DRAM Data	YD27		Hi-Z	SB
D21	YD28	IO	Y-BUS DDR-DRAM Data	YD28		Hi-Z	SB
C22	YD29	IO	Y-BUS DDR-DRAM Data	YD29		Hi-Z	SB
C21	YD30	IO	Y-BUS DDR-DRAM Data	YD30		Hi-Z	SB
B22	YD31	IO	Y-BUS DDR-DRAM Data	YD31		Hi-Z	SB
Y8	ZA0	O	Z-BUS DDR-DRAM Address	ZA0		F1	SO
V8	ZA1	O	Z-BUS DDR-DRAM Address	ZA1		F1	SO
W8	ZA2	O	Z-BUS DDR-DRAM Address	ZA2		F1	SO
AB3	ZA3	O	Z-BUS DDR-DRAM Address	ZA3		F1	SO
W16	ZA4	O	Z-BUS DDR-DRAM Address	ZA4		F1	SO
U15	ZA5	O	Z-BUS DDR-DRAM Address	ZA5		F1	SO
W15	ZA6	O	Z-BUS DDR-DRAM Address	ZA6		F1	SO
W14	ZA7	O	Z-BUS DDR-DRAM Address	ZA7		F1	SO
U14	ZA8	O	Z-BUS DDR-DRAM Address	ZA8		F1	SO
U13	ZA9	O	Z-BUS DDR-DRAM Address	ZA9		F1	SO
V9	ZA10	O	Z-BUS DDR-DRAM Address	ZA10		F1	SO
Y14	ZA11	O	Z-BUS DDR-DRAM Address	ZA11		F1	SO
V13	ZA12	O	Z-BUS DDR-DRAM Address	ZA12		F1	SO
Y9	ZBA0	O	Z-BUS DDR-DRAM Bank Address	ZBA0		F1	SO
U9	ZBA1	O	Z-BUS DDR-DRAM Bank Address	ZBA1		F1	SO
U12	ZCK	O	Z-BUS DDR-DRAM Clock	ZCK		F1	SD
W13	ZnCK	O	Z-BUS DDR-DRAM Inverting Clock	ZnCK		F1	SD
V12	ZCKE	O	Z-BUS DDR-DRAM Clock Enable	ZCKE		F1	SO
Y10	ZnCS0	O	Z-BUS DDR-DRAM Chip Select	ZnCS0		F1	SO
V10	ZnCS1	O	Z-BUS DDR-DRAM Chip Select	ZnCS1		F1	SO
W10	ZnRAS	O	Z-BUS DDR-DRAM RAS	ZnRAS		F1	SO
W11	ZnCAs	O	Z-BUS DDR-DRAM CAS	ZnCAs		F1	SO
U11	ZnWE	O	Z-BUS DDR-DRAM Write Enable	ZnWE		F1	SO
Y7	ZLDM0	O	Z-BUS Data Mask for ZD[7:0]	ZLDM0		F1	SO
Y12	ZUDM0	O	Z-BUS Data Mask for ZD[15:8]	ZUDM0		F1	SO
Y11	ZLDM1	O	Z-BUS Data Mask for ZD[23:16]	ZLDM1		F1	SO
V15	ZUDM1	O	Z-BUS Data Mask for ZD[31:24]	ZUDM1		F1	SO
Y6	ZLDQS0	IO	Z-BUS Data Strobe for ZD[7:0]	ZLDQS0		F1	SB
V11	ZUDQS0	IO	Z-BUS Data Strobe for ZD[15:8]	ZUDQS0		F1	SB
T11	ZLDQS1	IO	Z-BUS Data Strobe for ZD[23:16]	ZLDQS1		F1	SB
Y16	ZUDQS1	IO	Z-BUS Data Strobe for ZD[31:24]	ZUDQS1		F1	SB
AA4	ZD0	IO	Z-BUS DDR-DRAM Data	ZD0		F1	SB
AB4	ZD1	IO	Z-BUS DDR-DRAM Data	ZD1		F1	SB
AA5	ZD2	IO	Z-BUS DDR-DRAM Data	ZD2		F1	SB
AB5	ZD3	IO	Z-BUS DDR-DRAM Data	ZD3		F1	SB
AA6	ZD4	IO	Z-BUS DDR-DRAM Data	ZD4		F1	SB
AB6	ZD5	IO	Z-BUS DDR-DRAM Data	ZD5		F1	SB
AA7	ZD6	IO	Z-BUS DDR-DRAM Data	ZD6		F1	SB
AB7	ZD7	IO	Z-BUS DDR-DRAM Data	ZD7		F1	SB

AB8	ZD8	IO	Z-BUS DDR-DRAM Data	ZD8		F1	SB
AA8	ZD9	IO	Z-BUS DDR-DRAM Data	ZD9		F1	SB
AB9	ZD10	IO	Z-BUS DDR-DRAM Data	ZD10		F1	SB
AA9	ZD11	IO	Z-BUS DDR-DRAM Data	ZD11		F1	SB
AB10	ZD12	IO	Z-BUS DDR-DRAM Data	ZD12		F1	SB
AA10	ZD13	IO	Z-BUS DDR-DRAM Data	ZD13		F1	SB
AB11	ZD14	IO	Z-BUS DDR-DRAM Data	ZD14		F1	SB
AA11	ZD15	IO	Z-BUS DDR-DRAM Data	ZD15		F1	SB
AB12	ZD16	IO	Z-BUS DDR-DRAM Data	ZD16		F1	SB
AA12	ZD17	IO	Z-BUS DDR-DRAM Data	ZD17		F1	SB
AB13	ZD18	IO	Z-BUS DDR-DRAM Data	ZD18		F1	SB
AA13	ZD19	IO	Z-BUS DDR-DRAM Data	ZD19		F1	SB
AB14	ZD20	IO	Z-BUS DDR-DRAM Data	ZD20		F1	SB
AA14	ZD21	IO	Z-BUS DDR-DRAM Data	ZD21		F1	SB
AB15	ZD22	IO	Z-BUS DDR-DRAM Data	ZD22		F1	SB
AA15	ZD23	IO	Z-BUS DDR-DRAM Data	ZD23		F1	SB
AB16	ZD24	IO	Z-BUS DDR-DRAM Data	ZD24		F1	SB
AA16	ZD25	IO	Z-BUS DDR-DRAM Data	ZD25		F1	SB
AB17	ZD26	IO	Z-BUS DDR-DRAM Data	ZD26		F1	SB
AA17	ZD27	IO	Z-BUS DDR-DRAM Data	ZD27		F1	SB
AB18	ZD28	IO	Z-BUS DDR-DRAM Data	ZD28		F1	SB
AA18	ZD29	IO	Z-BUS DDR-DRAM Data	ZD29		F1	SB
AB19	ZD30	IO	Z-BUS DDR-DRAM Data	ZD30		F1	SB
AA19	ZD31	IO	Z-BUS DDR-DRAM Data	ZD31		F1	SB
Y1	WA0_WnDQM0 /GPIOE0	IO	W-BUS Address & ByteEnb[0] /GPIO : WA[25:0] outputs the memory address of the corresponding static banks. If the bus width of static bank is 16bit. Especially WA0 is used to be Lower byte enable.	WA0_WnDQM0		Note1	PB12
V2	WA1/GPIOE1	IO	W-BUS Address / GPIO	WA1		Note1	PB12
W1	WA2/GPIOE2	IO	W-BUS Address / GPIO	WA2		Note1	PB12
R5	WA3/GPIOE3	IO	W-BUS Address / GPIO	WA3		Note1	PB12
R2	WA4/GPIOE4	IO	W-BUS Address / GPIO	WA4		Note1	PB12
P5	WA5/GPIOE5	IO	W-BUS Address / GPIO	WA5		Note1	PB12
R1	WA6/GPIOE6	IO	W-BUS Address / GPIO	WA6		Note1	PB12
P2	WA7/GPIOE7	IO	W-BUS Address / GPIO	WA7		Note1	PB12
N3	WA8/GPIOE8	IO	W-BUS Address / GPIO	WA8		Note1	PB12
N2	WA9/GPIOE9	IO	W-BUS Address / GPIO	WA9		Note1	PB12
M4	WA10/GPIOE10	IO	W-BUS Address / GPIO	WA10		Note1	PB12
L1	WA11/GPIOE11	IO	W-BUS Address / GPIO	WA11		Note1	PB12
K1	WA12/GPIOE12	IO	W-BUS Address / GPIO	WA12		Note1	PB12
K2	WA13/WDREQ1 /GPIOE13	IO	W-BUS Address / DMA Request #1 / GPIO WA[13] can be used to be external DMA Request. Refer to memory controller part.	WA13	WDREQ1	Note1	PB12
H1	WA14/WDACK1 /GPIOE14	IO	W-BUS Address / DMA Acknowledge #1 / GPIO WA[14] can be used to be external DMA acknowledge	WA14	WDACK1	Note1	PB12
J2	WA15/IVSYNC /GPIOE15	IO	W-BUS Address / ISP Vertical Sync Input / GPIO WA[25:15] can be used for capturing external video.	WA15	IVSYNC	Note1	PB12
H2	WA16/IHSYNC/GPIOF0	IO	W-BUS Address / ISP Horizontal Sync Input / GPIO	WA16	IHSYNC	Note1	PB12
G2	WA17/ICLK/GPIOF1	O	W-BUS Address / ISP Video Input,Output Clock / GPIO	WA17	ICLK	Note1	PB12

			If WA[17] is used to be video capture clock, ICLK can be video input clock and video output clock				
J4	WA18/ID0/GPIOF2	O	W-BUS Address / ISP Video Input Data / GPIO ID[7:0], WA[25:18] can be used to be video input data bus.	WA18	ID0	Note1	PB12
H3	WA19/ID1/GPIOF3	O	W-BUS Address / ISP Video Input Data / GPIO	WA19	ID1	Note1	PB12
T1	WA20/ID2/GPIOF4	IO	W-BUS Address / ISP Video Input Data / GPIO	WA20	ID2	Note1	PB12
P4	WA21/ID3/GPIOF5	IO	W-BUS Address / ISP Video Input Data / GPIO	WA21	ID3	Note1	PB12
N5	WA22/ID4/GPIOF6	IO	W-BUS Address / ISP Video Input Data / GPIO	WA22	ID4	Note1	PB12
N4	WA23/ID5/GPIOF7	IO	W-BUS Address / ISP Video Input Data / GPIO	WA23	ID5	Note1	PB12
M3	WA24/ID6/GPIOF8	IO	W-BUS Address / ISP Video Input Data / GPIO	WA24	ID6	Note1	PB12
M1	WA25/ID7/GPIOF9	IO	W-BUS Address / ISP Video Input Data / GPIO	WA25	ID7	Note1	PB12
V5	WD0/GPIOG0	IO	W-BUS Data / GPIO	WD0		Note1	PB12
W4	WD1/GPIOG1	IO	W-BUS Data / GPIO	WD1		Note1	PB12
V4	WD2/GPIOG2	IO	W-BUS Data / GPIO	WD2		Note1	PB12
U5	WD3/GPIOG3	IO	W-BUS Data / GPIO	WD3		Note1	PB12
Y2	WD4/GPIOG4	IO	W-BUS Data / GPIO	WD4		Note1	PB12
AA1	WD5/GPIOG5	IO	W-BUS Data / GPIO	WD5		Note1	PB12
T5	WD6/GPIOG6	IO	W-BUS Data / GPIO	WD6		Note1	PB12
W2	WD7/GPIOG7	IO	W-BUS Data / GPIO	WD7		Note1	PB12
Y3	WD8/VD16/GPIOG8	IO	W-BUS Data / Video Output Data /GPIO If 24bit digital video output is required. WD[15:8] can be used to be Upper byte (VD[23:16]).	WD8	VD16	Note1	PB12
W3	WD9/VD17/GPIOG9	IO	W-BUS Data / Video Output Data /GPIO	WD9	VD17	Note1	PB12
V3	WD10/VD18/GPIOG10	IO	W-BUS Data / Video Output Data /GPIO	WD10	VD18	Note1	PB12
U4	WD11/VD19/GPIOG11	IO	W-BUS Data / Video Output Data /GPIO	WD11	VD19	Note1	PB12
U3	WD12/VD20/GPIOG12	IO	W-BUS Data / Video Output Data /GPIO	WD12	VD20	Note1	PB12
T4	WD13/VD21/GPIOG13	IO	W-BUS Data / Video Output Data /GPIO	WD13	VD21	Note1	PB12
U2	WD14/VD22/GPIOG14	IO	W-BUS Data / Video Output Data /GPIO	WD14	VD22	Note1	PB12
T2	WD15/VD23/GPIOG15	IO	W-BUS Data / Video Output Data /GPIO	WD15	VD23	Note1	PB12
C5	WnCS0/WnNCS6	O	Static Memory Bank0 Chip Select / NAND Bank6 It is used to be Static memory bank0 chip select or chip select for NAND bank6. its function is selected by cpu, Reset default function is WnCS0 If ROM/NOR flash booting is required. The chip selection of boot memory is connected to it.	WnCS0_WnNCS6		F1	P8
B4	WnCS1/WnNCS7	O	Static Memory Bank1 Chip Select / NAND Bank7	WnCS1_WnNCS7		F1	P8
B3	WnNCS0/WnCS2	O	NAND Bank0 / Static Memory Bank2 Chip Select If NAND Boot is required. The Chip selection of NAND Boot memory must to be connected to it.	WnNCS0_WnCS2		F1	P8
A2	WnNCS1/WnCS3	O	NAND Bank1 / Static Memory Bank3 Chip Select	WnNCS1_WnCS3		F1	P8
E6	WnPCS0/WnCS4/WnNCS2	O	PCMCIA Chip Select 0 / Static Bank4 / NAND Bank2	WnPCS0_WnCS4_WnNCS2		F1	P8
E7	WnPCS1/WnCS5/WnNCS3	O	PCMCIA Chip Select 1 / Static Bank5 / NAND Bank3	WnPCS1_WnCS5_WnNCS3		F1	P8
B7	WnICS0/WnNCS4	O	IDE Controller Chip select /NAND Chip Select Bank4 In IDE Chip select, It is used to select IDE card, and enable the High byte lane.	WnICS0_WnNCS4		F1	P8
C7	WnICS1/WnNCS5	O	IDE Controller Chip select /NAND Chip Select Bank5 In IDE Chip select, It is used to select IDE card, and enable the Low byte lane.	WnICS1_WnNCS5		F1	P8
A3	WnWE	O	Static/PCMCIA/IDE Write Enable	WnWE		F1	P8

C6	WnOE	O	Static/PCMCIA/IDE Read Enable	WnOE		F1	P8
F6	WnDQM1	O	High Byte Enable, ByteEnb[1]	WnDQM1		F1	P8
D6	WnNOE	O	NAND Read Enable	WnNOE		F1	P8
A4	WnNWE	O	NAND Write Enable	WnNWE		F1	P8
B6	WRDnWR	O	Buffer Direction Control if static memory is connected through buffer, It is used to be buffer controller signal of external buffer.	WRDnWR		F1	P8
P3	WnPior_CLE/GPIOH0	IO	PCMCIA,IDE I/O Read Enable/ NAND CLE (Command Latch Enable) / GPIO	WnPior_CLE		F1	PB12
P1	WnPIOW_ALE/GPIOH1	IO	PCMCIA,IDE I/O Read Enable / NAND ALE (Address Latch Enable) / GPIO	WnPIOW_ALE		F1	PB12
M5	WSKTSEL/GPIOH2	IO	PCMCIA Socket Select / GPIO It is an output and is used by external steering logic to route control, address and data signals to one of the two PCMCIA sockets	WSKTSEL		F1	PB12
N1	WnPREG/GPIOH3	IO	PCMCIA nPREG / GPIO This signal is an output and indicates that, on a memory transaction, the target address is attribute space. This signal has same timing as address	WnPREG		F1	PB12
U1	WnIOIS16/GPIOH4	IO	PCMCIA IOIS16 Signal / GPIO Is is an input and is an acknowledgement from the PCMCIA card that the current address is a valid 16bit wide I/O address	WnIOIS16		F1	PB12
R3	WRnB/GPIOH5	IO	Static/NAND/IDE/PCMCIA Wait Control / GPIO Is is an input and is driven low by the PCMCIA/CF/IDE/Static Memory to extend the length of the transfers to/from application processor	WRnB		F1	PB12
L5	WDREQ0/GPIOH6	IO	External DMA Request / GPIO	WDREQ0		F1	PB12
K5	WDACK0/GPIOH7	IO	External DMA Acknowledge / GPIO	WDACK0		F1	PB12
B5	EIRQ0	IO	External Interrupt 0	EIRQ0		F1	PI
A6	EIRQ1	IO	External Interrupt 1	EIRQ1		F1	PI
F1	EIRQ2/GPIOK12	IO	External Interrupt 2 / GPIO	EIRQ2		GI	PB12
H5	EIRQ3/CTS1/GPIOK13	IO	External Interrupt 3 / UART1 CTS / GPIO	EIRQ3	CTS1	GI	PB12
W5	USBD0+	IO	USB0 Port : D+ : USB Host Data port	USBD+		F1	US
AB2	USBD0-	IO	USB0 Port : D- : USB Host Data port	USBD-		F1	US
AB1	USBD1+	IO	USB1 Port : D+ : USB Host Data port / USB Device Data port It can be used to be USB Host data or USB Device data, Its function is decided by cpu, refer to GPIO Misc. register.	USBD+		F1	US
AA2	USBD1-	IO	USB1 Port : D- : USB Host Data port / USB Device Data port	USBD-		F1	US
U6	SCL	IO	I2C Clock	SCL		F1	OD
T3	SDA	IO	I2C Data	SDA		F1	OD
K3	SDDAT0/ GPIOH8	IO	MMC-SD Data bit 0 / GPIO	SDDAT0		GI	PB12
L3	SDDAT1/ GPIOH9	IO	MMC-SD Data bit 1 / GPIO	SDDAT1		GI	PB12
J1	SDDAT2/GPIOH10	IO	MMC-SD Data bit 2 / GPIO	SDDAT2		GI	PB12
L4	SDDAT3/GPIOH11	IO	MMC-SD Data bit 3 / GPIO	SDDAT3		GI	PB12
L2	SDCLK/GPIOH12	O	MMC-SD Clock / GPIO	SDCLK		GI	PB12
M2	SDCMD/ GPIOH13	IO	MMC-SD Command / GPIO	SDCMD		GI	PB12
V6	PWM0/RTS1/GPIOH14	IO	PWM0 Output / UART1 RTS / GPIO	PWM0	RTS1	GI	PB12
Y4	PWM1/GPIOH15	IO	PWM1 Output / GPIO	PWM1		GI	PB12
C3	ADCIN0	AI	ADC Channel 0 Input	ADCIN0		F1	GP
D4	ADCIN1	AI	ADC Channel 1 Input	ADCIN1		F1	GP
E5	ADCIN2	AI	ADC Channel 2 Input	ADCIN2		F1	GP
F4	ADCIN3	AI	ADC Channel 3 Input	ADCIN3		F1	GP
F5	AREF	AI	ADC Voltage Reference	AREF		F1	GP
E20	NGPIO0/DTR1/ID0	IO	NGPIO / UART1 DTR / Video In / GPIO	DTR1	ID0	GI	PB12

	/GPIO0		NGPIO[15:0] : When interfacing with external CPU, Its bus is used to get the data which external cpu send to VR3D. ID[7:0] : When capturing the video, Its bus is used as Video input data bus.				
G19	NGPIO1/DSR1/ID1 /GPIO1	IO	NGPIO / UART1 DSR / Video In / GPIO	DSR1	ID1	GI	PB12
A22	NGPIO2/DCD1/ID2 /GPIO2	IO	NGPIO / UART1 DCD / Video In / GPIO	DCD1	ID2	GI	PB12
F19	NGPIO3/RTS1/ID3 /GPIO3	IO	NGPIO / UART1 RTS / Video In / GPIO	RTS1	ID3	GI	PB12
D20	NGPIO4/CTS1/ID4 /GPIO4	IO	NGPIO / UART1 CTS / Video In / GPIO	CTS1	ID4	GI	PB12
E18	NGPIO5/RI1/ID5 /GPIO5	IO	NGPIO / UART1 RI / Video In / GPIO	RI1	ID5	GI	PB12
E19	NGPIO6/ID6/GPIO6	IO	NGPIO / Video In / GPIO		ID6	GI	PB12
D19	NGPIO7/ID7/GPIO7	IO	NGPIO / Video In / GPIO		ID7	GI	PB12
C20	NGPIO8/DTR0/IVSYNC /GPIO8	IO	NGPIO / UART0 DTR / Video In Vertical Sync / GPIO	DTR0	IVSYNC	GI	PB12
C19	NGPIO9/DSR0/IHSYNC /GPIO9	IO	NGPIO / UART0 DSR / Video In Horizontal Sync / GPIO	DSR0	IHSYNC	GI	PB12
D17	NGPIO10/DCD0/ICLK /GPIO10	IO	NGPIO / UART0 DCD / Video In Clock / GPIO	DCD0	ICLK	GI	PB12
B21	NGPIO11/RTS0 /SSPRXD/GPIO11	IO	NGPIO / UART0 RTS / SSP Receiver Port / GPIO	RTS0	SSPRXD	GI	PB12
D18	NGPIO12/CTS0 /SSPCLK/GPIO12	IO	NGPIO / UART0 CTS / SSP Clock / GPIO	CTS0	SSPCLK	GI	PB12
G3	NGPIO13/RI0 /SSPFRM/GPIO13	IO	NGPIO / UART0 RI / SSP FRM / GPIO	RI0	SSPFRM	GI	PB12
E1	NGPIO14/RX1 /GPIO14	IO	NGPIO / UART1 RX / GPIO	RX1		GI	PB12
F2	NGPIO15/TX1/GPIO15	IO	NGPIO / UART1 TX / GPIO	TX1		GI	PB12
V1	NGPIOLAT/SSPTXD /GPIOK0	IO	GPIO Latch Enable / SSP Transmitter port / GPIO	NGPIOLAT	SSPTXD	GI	PB12
H4	ABCK/BCK/GPIOK1	IO	AC97 BCK / I2S BCK / GPIO	ABCK	BCK	GI	PB12
G1	ASYNC/LRCK/GPIOK2	IO	AC97 Sync / I2S LRCK / GPIO	ASYNC	LRCK	GI	PB12
J3	ADATIN/SDAT2 /GPIOK3	IO	AC97 Data In / I2S Data Channel 2 / GPIO	ADATIN	SDAT2	GI	PB12
G4	nARST/SDAT1/GPIOK4	IO	AC97 Reset / I2S Data Channel 1 / GPIO	nARST	SDAT1	GI	PB12
K4	ADATOUT/SDAT0/GPIOK5	IO	AC97 Data Output / I2S Data Channel 2 / GPIO	ADATOUT	SDAT0	GI	PB12
V16	VD0/GPIOJ0/CFG0	IO	Digital Video Data Output / GPIO / CfgShadow VD[15:0] is Digital video output bus, It is used as reset configuration parameter to determine operation mode. It must be tied to GND to VDD(3.3v) through pull-up/pull-down resistor. CfgShadow : It determine to enable/disable memory shadow. 0 : Non Shadow (Base 0 : Bank-W), Refer to memory controller. 1 : Shadow (Base 0 : Bank-z)		VD0	GI	PB12
Y18	VD1/GPIOJ1/CFG1	IO	Digital Video Data Output / GPIO / CfgBootType[0] CfgBootType[1:0] : It determines the booting method. 00 : ROM/NOR Flash 10 : NAND Boot		VD1	GI	PB12
W18	VD2/GPIOJ2/CFG2	IO	Digital Video Data Output / GPIO / CfgBootType[1]		VD2	GI	PB12
AA20	VD3/GPIOJ3/CFG3	IO	Digital Video Data Output / GPIO / CfgNFBSIZE, CfgSDRLATx CfgNFBSIZE : Reserved, Its name and function is changed to CfgSDRLATx in VR3Dr2 CfgSDRLATx : Its determines the CAS latency of DDR-DRAM 0 : 2 Cycle 1 : 3 Cycle		VD3	GI	PB12
AB21	VD4/GPIOJ4/CFG4	IO	Digital Video Data Output / GPIO / CfgNFBUSWidth		VD4	GI	PB12

			CfgNFBUSWidth : It determines the bus width of NAND Flash which is used as NAND booting memory. 0 : 8bit NAND Flash 1 : 16bit NAND Flash				
Y19	VD5/GPIOJ5/CFG5	IO	Digital Video Data Output / GPIO / CfgNFTType CfgNFTType : It determines the NAND Address Type, 0 : 3 address type 1 : 4 address type		VD5	GI	PB12
V17	VD6/GPIOJ6/CFG6	IO	Digital Video Data Output / GPIO / CfgSR0BUSWidth CfgSR0BUSWidth : It determines the bus width of Static memory (ROM/NOR flash) which is controlled by nSCS0 0 : 8bit 1 : 16bit		VD6	GI	PB12
AB22	VD7/GPIOJ7/CFG7	IO	Digital Video Data Output / GPIO / CfgSDRTYPE CfgSDRTYPE : Reserved : It must be tied to VDD through pull-up resistor		VD7	GI	PB12
V18	VD8/GPIOJ8/CFG8	IO	Digital Video Data Output/ GPIO / CfgSDRBUSBW[0] CfgSDRBUSBW[1:0] : It determines the bus width of Bank-Z 10 : 16bit 11 : 32bit If CfgShadow is 1, The Reset Configuration be related to DDR-DRAM must be set.		VD8	GI	PB12
AA21	VD9/GPIOJ9/CFG9	IO	Digital Video Data Output / GPIO / CfgSDRBUSBW[1]		VD9	GI	PB12
W19	VD10/GPIOJ10/CFG10	IO	Digital Video Data Output / GPIO / CfgSDRBW[0] CfgSDRBW[1:0] : It determines the data bit width of a DDR-DRAM component. 00 : Reserved 01 : 8bit 10 : 16bit 11 : 32bit		VD10	GI	PB12
Y20	VD11/GPIOJ11/CFG11	IO	Digital Video Data Output / GPIO / CfgSDRBW[1]		VD11	GI	PB12
Y21	VD12/GPIOJ12/CFG12	IO	Digital Video Data Output / GPIO / CfgSDRCAP[0] CfgSDRCAP[1:0] : It determines the capacity of a DDR-DRAM component. 00 : 64Mbit 01 : 128Mbit 10 : 256Mbit 11 : 512Mbit		VD12	GI	PB12
W20	VD13/GPIOJ13/CFG13	IO	Digital Video Data Output / GPIO / CfgSDRCAP[1]		VD13	GI	PB12
AA22	VD14/GPIOJ14	IO	Digital Video Data Output / GPIO		VD14	GI	PB12
V19	VD15/GPIOJ15	IO	Digital Video Data Output / GPIO		VD15	GI	PB12
W17	VSYNCGPIOK6	IO	Vertical Sync Output / GPIO	VSYNCGPIOK6		GI	PB12
AB20	HSYNCGPIOK7	IO	Horizontal Sync Output / GPIO	HSYNCGPIOK7		GI	PB12
W6	CSYNCGPIOK8	IO	Composite Sync Output / GPIO	CSYNCGPIOK8		GI	PB12
V20	VCLKGPIOK9	IO	Video Clock Input/Output / GPIO	VCLKGPIOK9		GI	PB12
F3	RX0GPIOK10	IO	UART0 RX / GPIO	RX0GPIOK10		F1	PB12
G5	TX0GPIOK11	IO	UART0 TX / GPIO	TX0GPIOK11		F1	PB12
C4	ADCVDD3D	P	3.3V for ADC Digital Block				
E4	ADCVDDA	P	3.3V for ADC Analog Block				
D5	ADCVSS3D	P	Ground for ADC Digital Block				
E3	ADCVSSA	P	Ground for ADC Analog Block				
B2	PLLVDDA	P	1.8V for A-PLL				
D2	PLLVDDF	P	1.8V for F-PLL				
D3	PLLVDDU	P	1.8V for U-PLL				
C2	PLLVSSA	P	Ground for A-PLL				
E2	PLLVSSF	P	Ground for F-PLL				
A1	PLLVSSU	P	Ground for U-PLL				
A5	VDD3A	P	3.3V for I/O PAD				
F9, F10, F16, G18, J6, L6, P6, T18, U17, Y5, Y17, D10	VDD1	P	1.8V for Core				
AA3, F18, J5, M6, R4, U19	VDD3	P	3.3V for I/O PAD				
E17, K6, U18	VDD1A	P	1.8V for Core				

A11	XREF	P	Bank X Voltage Reference, DDR:1.25V		
L20	YREF	P	Bank Y Voltage Reference, DDR:1.25V		
W12	ZREF	P	Bank Z Voltage Reference, DDR:1.25V		
C18	VDD1X	P	Bank X 1.8V		
J17	VDD1Y	P	Bank X 1.8V		
W9	VDD1Z	P	Bank X 1.8V		
B12, C15, F11	VDD2X	P	Bank X PAD Power, DDR:2.5V		
H18, J18, K19, M20, P17, R18	VDD2Y	P	Bank Y PAD Power, DDR:2.5V		
U10, V7, V14, W7, Y13, Y15	VDD2Z	P	Bank Z PAD Power, DDR:2.5V		
F7, F8, F15, F17G6, G10, G11, G12, G13, G17, H6, H17, K7, K16, L7, L16, M7, M16, N6, N7, N16, N17, R6, R17, T6, T10, T12, T13, T17, U7, U8, U16	GND	P	Ground		

I/O Type	Description
PB8	Bi-Directional pad, LVCMOS Schmitt-trigger, 100Kohm pull-up resistor with control, tri-state, Io=8mA
PB12	Bi-Directional pad, LVCMOS Schmitt-trigger, 100Kohm pull-up resistor with control, tri-state, Io=12mA
PI	Input pad, LVCMOS Schmitt-trigger
M26	Oscillator cell with enable and feedback resistor
OD	Open Drain PAD
GP	Analog PAD
SO	SSTL output pad, Io=24mA
SD	SSTL Differential pad, Io=24mA
SB	SSTL Bi-Directional pad, Io=24mA
US	USB Pad

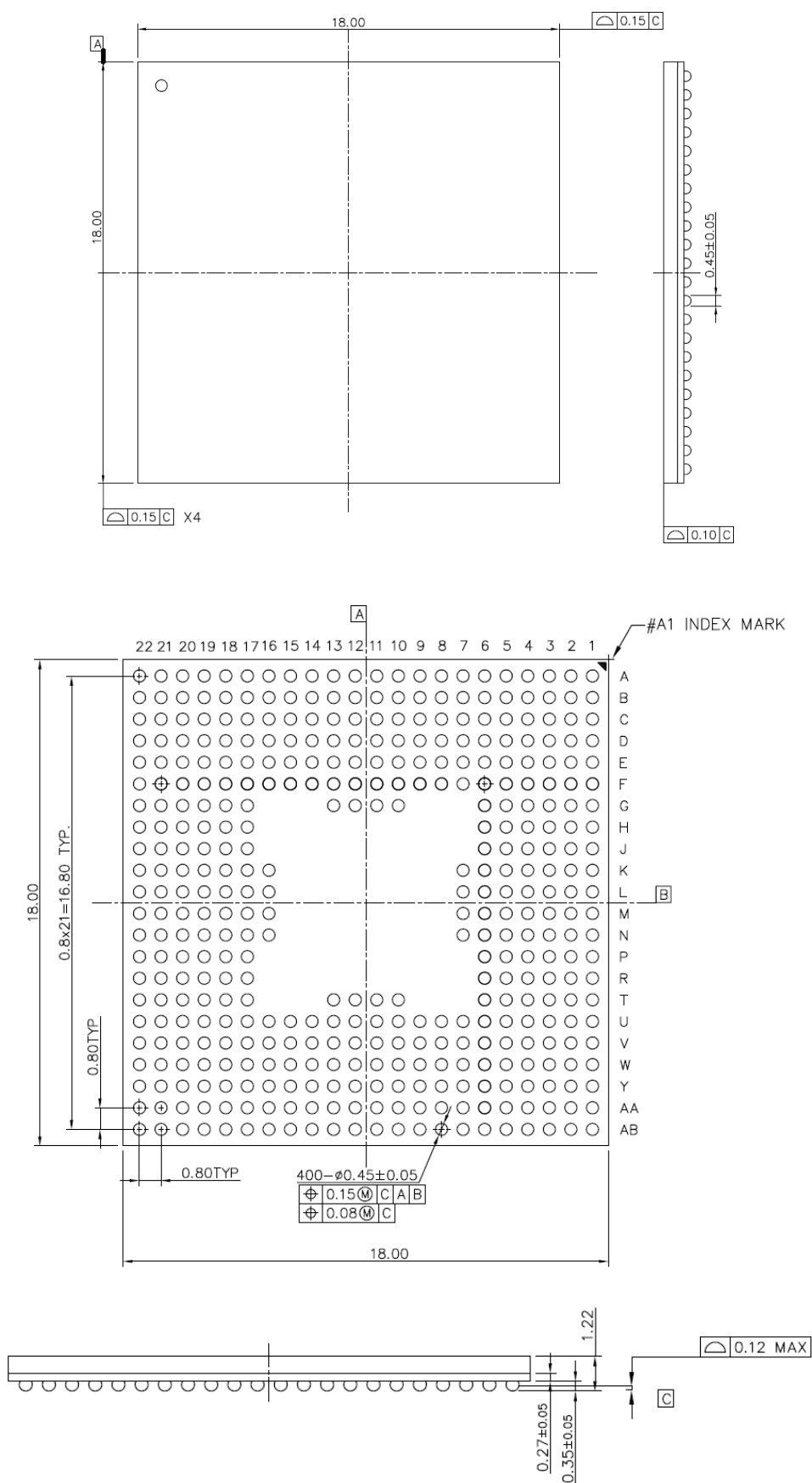
Note1 : It is determined by reset configuration, Refer to memory controller part

2.2 Ball Location

	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	
A	NGPIO2	XD0	XD2	XD4	XD6	XA10	XD8	XD10	XD12	XD14	XA9	XREF	PWRXTO	PWRXTI	TESTEN	TDO	EIRQ1	VDD3	WnNWE	WnWE	WnNCS1	PLLVS	A
B	YD31	NGPIO1	XD1	XD3	XD5	XBA0	XD7	XD9	XD11	XD13	VDD2x	XD15	XnCK	TMS	nVDDF	WnICS0	WRDnWR	EIRQ0	WnCS1	WnNCS0	PLLVD	XTALO	B
C	YD29	YD30	NGPIO8	NGPIO9	VDD1	XLDQS	XnWE	VDD2x	XA2	XA3	XA7	XCKE	XCK	TCK	PWRON	WnICS1	WnOE	WnCS0	ADC VDD3D	ADCIN0	PLLVS	XTALI	C
D	YD27	YD28	NGPIO4	NGPIO7	NGPIO12	NGPIO10	XnRAS	XnCS	XA1	XA5	XA12	XUDM	VDD1	TEST SCAN	TDI	nPORST	WnNOE	ADC VSS3D	ADCIN1	PLLVD	PLLVD	EXTCLK	D
E	YA6	YUDQS1	NGPIO0	NGPIO6	NGPIO5	VDD1	XLDM	XnCAS	XA0	XA4	XA8	XUDQS	nTRST	nBATF	nRSTOUT	WnPCS1	WnPCS0	ADCIN2	ADC VDDA	ADC VSSA	PLLVS	NGPIO14	E
F	YD25	YD26	YA4	NGPIO3	VDD3	GND	VDD1	GND	XBA1	XA6	XA11	VDD2x	VDD1	VDD1	GND	GND	WnDQM1	AREF	ADCIN3	RX0	NGPIO15	EIRQ2	F
G	YD24	YD23	YUDM1	NGPIO1	VDD1	GND				GND	GND	GND	GND				GND	TX0	nARST	NGPIO13	WA17	ASYN	G
H	YD22	YD21	YA8	YA5	VDD2y	GND											GND	EIRQ3	ABCK	WA19	WA16	WA14	H
J	YD20	YD19	YA7	YA9	VDD2y	VDD1											VDD1	VDD3	WA18	ADATIN	WA15	SDDAT2	J
K	YD18	YD17	YCK	VDD2y	YA12	YA11	GND									GND	VDD1	WDACK0	ADAT0	SDDAT0	WA13	WA12	K
L	YD16	YD15	YREF	YnCK	YUDQS0	YCKE	GND									GND	VDD1	WDREQ0	SDDAT3	SDDAT1	SDCLK	WA11	L
M	YD14	YD13	VDD2y	YnWE	YnRAS	YUDM0	GND									GND	VDD3	WSKTSEL	WA10	WA24	SDCMD	WA25	M
N	YD12	YD11	YnCAS	YLDM1	YBA0	GND	GND									GND	GND	WA22	WA23	WA8	WA9	WnPREG	N
P	YD10	YD9	YnCS	YLDQS1	YA0	VDD2y											VDD1	WA5	WA21	WnPIOR	WA7	WnPIOW	P
R	YD8	YD7	YA10	YBA1	VDD2y	GND											GND	WA3	VDD3	WRnB	WA4	WA6	R
T	YD6	YD5	YA2	YA3	VDD1	GND				GND	GND	ZLDQS1	GND				GND	WD6	WD13	SDA	WD15	WA20	T
U	YA1	YLDQS0	YLDM0	VDD3	VDD1	VDD1	GND	ZA5	ZA8	ZA9	ZCK	ZnWE	VDD2z	ZBA1	GND	GND	SCL	WD3	WD11	WD12	WD14	WnIOIS16	U
V	YD4	YD3	VCLK	VD15	VD8	VD6	VD0	ZUDM1	VDD2z	ZA12	ZCKE	ZUDQS0	ZnCS1	ZA10	ZA1	VDD2z	PWM0	WD0	WD2	WD10	WA1	NGPIO1AT	V
W	YD2	YD1	VD13	VD10	VD2	VSYN	ZA4	ZA6	ZA7	ZnCK	ZREF	ZnCAS	ZnRAS	VDD1	ZA2	VDD2z	CSYN	USB0+	WD1	WD9	WD7	WA2	W
Y	YD0	VD12	VD11	VD5	VD1	VDD1	ZUDQS1	VDD2z	ZA11	VDD2z	ZUDM0	ZLDM1	ZnCS0	ZBA0	ZA0	ZLDM0	ZLDQS0	VDD1	PWM1	WD8	WD4	WA0	Y
AA	VD14	VD9	VD3	ZD31	ZD29	ZD27	ZD25	ZD23	ZD21	ZD19	ZD17	ZD15	ZD13	ZD11	ZD9	ZD6	ZD4	ZD2	ZD0	VDD3	USB01-	WD5	AA
AB	VD7	VD4	HSYN	ZD30	ZD28	ZD26	ZD24	ZD22	ZD20	ZD18	ZD16	ZD14	ZD12	ZD10	ZD8	ZD7	ZD5	ZD3	ZD1	ZA3	USB00-	USB01+	AB
	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	

Bottom
View

2.3 Package Dimension



3. Power Management Controller

3.1 Overview

Power Management executes the below functions.

- Power Up Sequence
- Reset Generation
- Power Management
- Change PLL Value

PWRXTI clock (32.768Khz) is used as clock by power management.

The main functionality is Power up sequence control which make the VR3D to be stable state after power on, and latch the reset configuration value which determines the initial operation boot mode, etc in reset state.

This block makes a several reset signals such as External Reset Output (nRSTOUT), GPIO Reset, Soft Reset according to nPORST (Power On Reset PAD) and watchdog reset.

And VR3D has various power management schemes to keep optimal power consumption for a given task. The power management block consists of three power mode as below.

- Normal Mode
- IDLE Mode
- STOP Mode

IDLE mode : In this mode, All clocks are supplied to each block except for CPU (FCLK). The functionality of this mode is same as WaitForIRQ on ARM9. The IDLE mode results in reduced power consumption due to CPU core. Any interrupt request to CPU can be woken up from the IDLE mode. The cpu core define the interrupt source by setting interrupt enable bit in Interrupt controller. In addition to interrupt request generated by interrupt controller, The USB, Touch PAD are used as wakeup source.

STOP mode : In this mode, The Clock controller does not supply the clock for all block including ARM. The VR3D changes the state of DRAM to the self refresh mode before stopping the clock. In stop mode, The PLL, XTAL pad are changed to power down mode in order to achieve the low power consumption.

In STOP mode, the limited wakeup source (such as external interrupt, USB, TouchPAD) can be woken up from the STOP mode, because of The clock controller doest not supply the clock for all block except Power management.

4. Clock Controller

4.1 Overview

The Clock controller in VR3D can generate the required clock signals including FCLK for CPU, BCLK for Bus Controller, PCLK for peripheral interface, and various clock for peripheral operation. The VR3D has three internal PLL (FPLL, UPLL, APLL).

The Clock controller support dynamic power control to reduce power consumption in normal operation state. If the dynamic power control function is enabled, Each block enables or disables the its clock itself.

Figure 5.1 shows the internal clock path diagram of Clock controller.

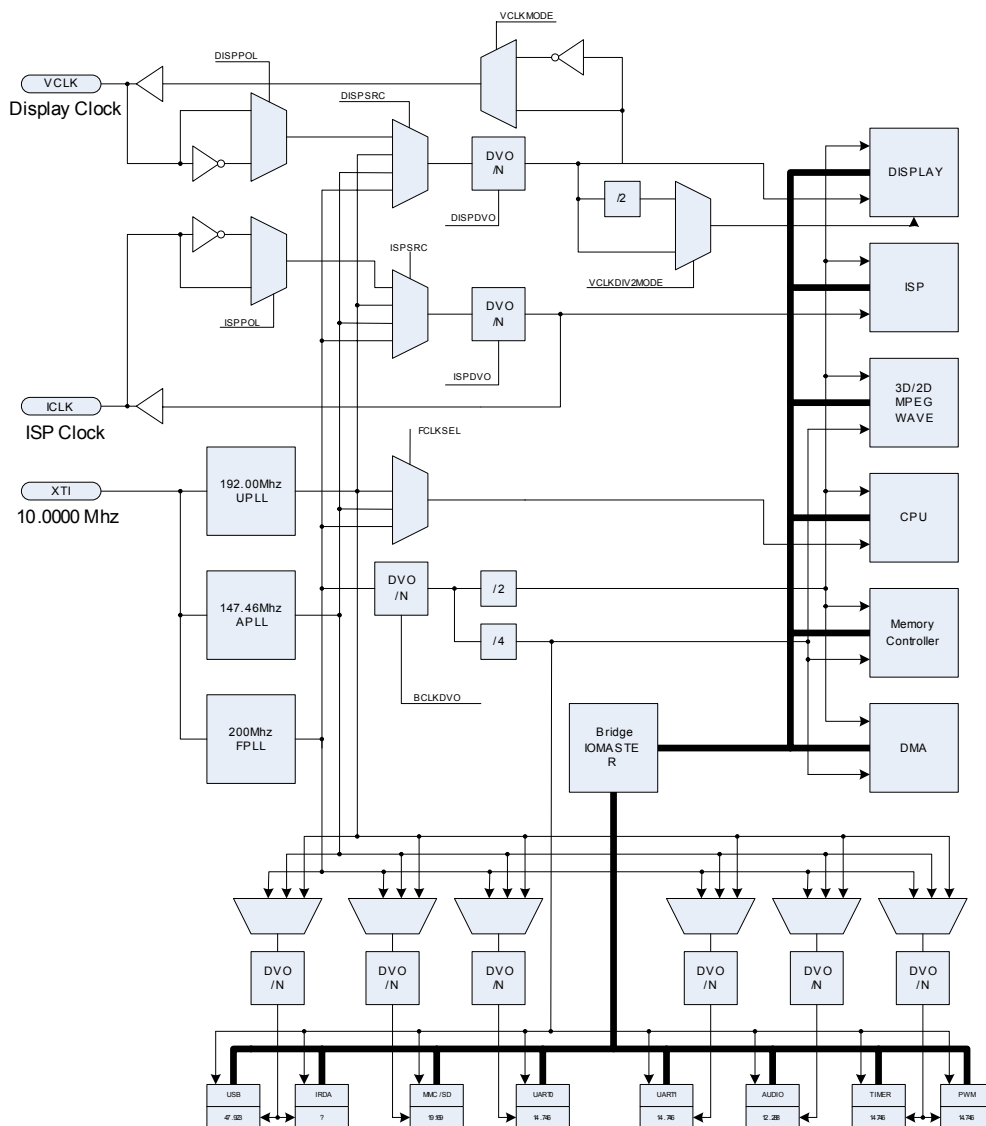


Figure 4.1 Clock Path diagram

The main clock source comes from an external crystal(XTI). The internal PLL generate the high frequency clock required in VR3D.

VR3D recommends that Input frequency of XTI is slower than 10.0000Mhz. If the input frequency is faster than 10.0000Mhz, VR3D would operate unstable because the BCLK (for Bus Controller, System Controller) is faster than 100Mhz in booting stage.

5. GPIO Controller

5.1 Overview

The VR3D has the ten GPIO Group, GPIOA~GPIOK except GPIOD. Each GPIO group is composed of several GPIO bit. Each GPIO bit can be GPIO output or GPIO In, In GPIO Input mode, GPIO Controller can detect the rising edge or falling edge or High or low level according to GPIO Detect Mode register (**GPIODET**). And The detected event is accessed by CPU or can activate the GPIO Interrupt to CPU. And Input signal through GPIO pad are sampled in PCLK(50Mhz) because of reducing the external noise on the signal.

Figure 5.1 shows the data path for a GPIO pad.

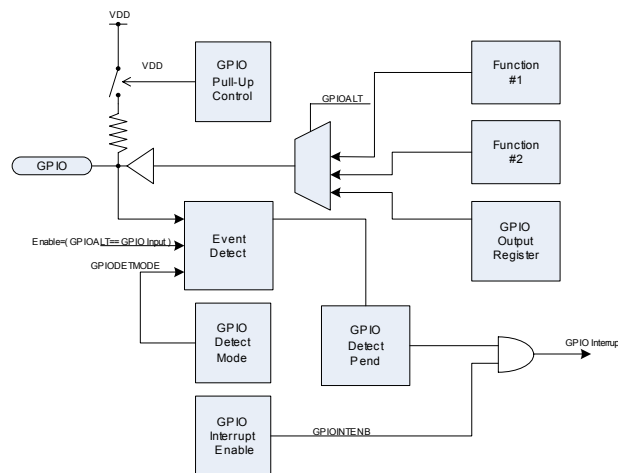


Figure 5.1 GPIO Data path block diagram

The GPIOA~GPIOC Pad are SSTL pad for DDR Memory, When these pad is used as input pad, These pad dissipates more power than normal pad and output mode. So, These pad has power down control when these pad are not used. The GPIOD~GPIOE has controllable Pull-Up Resistor .

GPIO Misc Register (**GPIONISC**) determines the wakeup enable of Touch PAD, USB, and the pad status of USB for detecting the usb event in power down mode, and the USB function selection (USB Host or USB device), determines the event detection mode for external Interrupt pad.

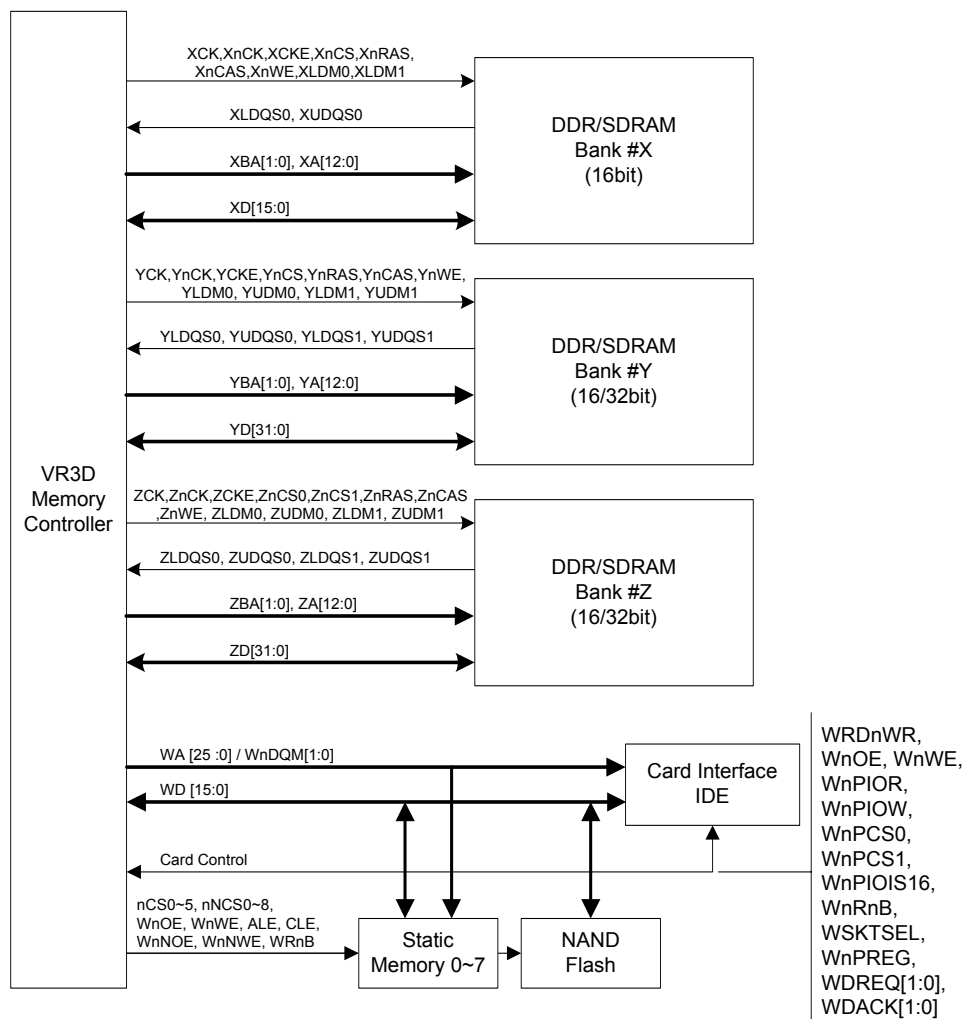
The USB1+/- Pads can be used as USB device pad or USB Host pad, The UDCSEL bit of GPIO Misc. Register(**GPIONISC**) determines what functionality.

6. Memory Controller

6.1 Overview

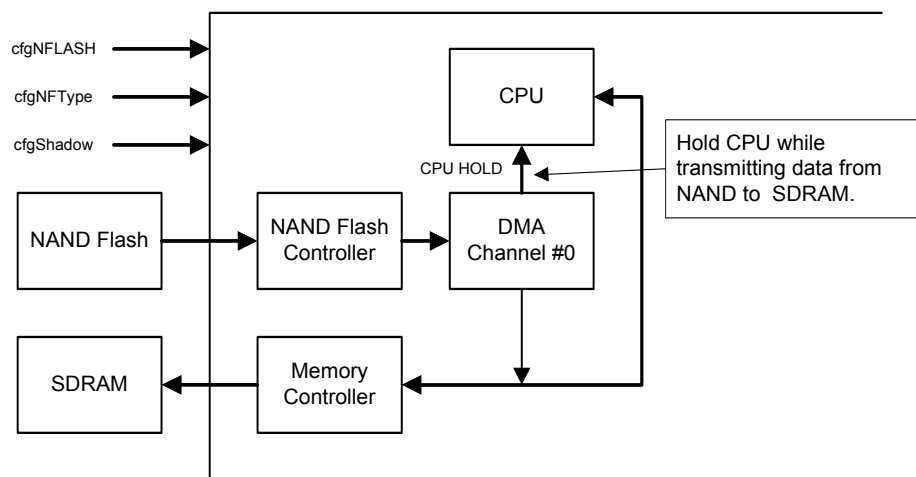
The VRender3D memory controller is based on a Unified Memory Architecture (UMA). This controller consists of three main control unit : dynamic memory, static memory, PCMCIA/CF interface. Each unit has dedicated control pins

- Unified Memory Architecture (UMA)
 - Three Banks of DDR-SDRAM (16 or 32bit data supported)
 - ◆ Support Active Power down mode automatically.
 - ◆ Support Self Refresh mode
 - ◆ Support three CAS Latency : 2, 2.5, 3 Cycles
 - Six Static Memory Chip Selects
 - Card interface supporting PCMCIA or Compact Flash
 - Supports UDMA(Ultra DMA), MDMA(Multiword DMA) for IDE
 - Four Separate Memory Controller (Bank#W, Bank #X , Bank #Y, Bank #Z)



6.2 NAND Boot Overview

VRender3D supports the booting function with NAND Flash.
The following figure shows the data path of booting with NAND.



FEATURE

1. Support 8 NAND Flash Banks. : x8, x16bit Bus Width
2. Contain Hardware ECC Block
3. Support NAND Flash Booting

NAND BOOT SEQUENCE

1. In case cfgNFBOOT is high, DMA prohibits main memory access as DMA holds CPU.
2. NAND Flash Controller which reads the data of external NAND flash saves at internal buffer.
3. NAND Flash requests DMA transmission to DMA Channel#0.
4. DMA transmits to main memory.
5. Execute 16/32KB repeatedly according to cfgNFSize.
6. If the transmission is completed, make CPU become normal operation as lifting CPU hold.

Caution when NAND Boot

- When NAND Boot, there should be Read/Writable Memory of 2/4KB in physical address 0 according to cfgNFSize.
- Because ECC is not executed when NAND boot, there should be no bit fail in the early 2/4Kbyte of flash.

7. Interrupt Controller

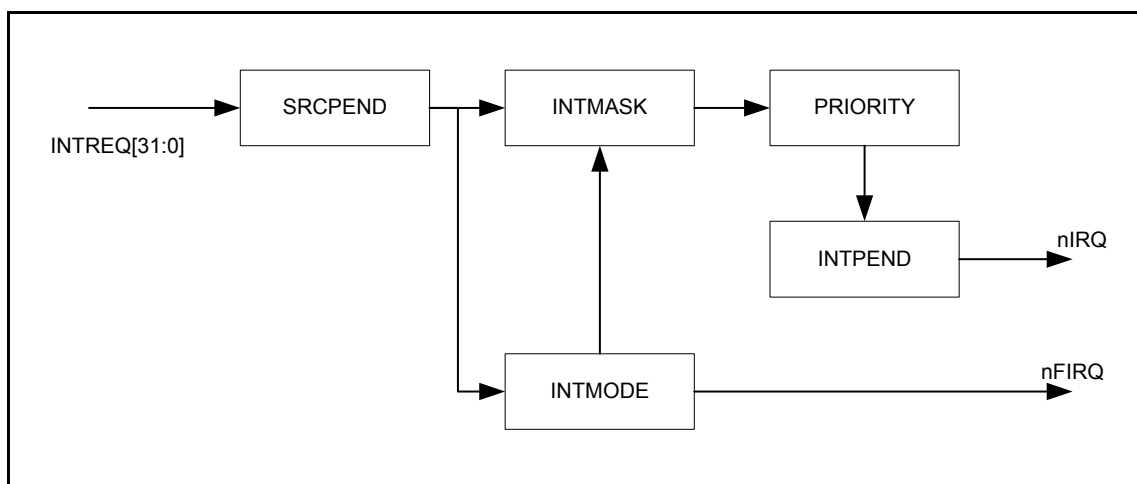
7.1 Overview

ARM920T in 28 interrupt sources from internal peripherals, including the DMA controller, UART, I2C and GPIO, etc. supply requests to an Interrupt Controller.

An FIQ or an IRQ (interrupt requests) are signaled by the Interrupt Controller to ARM920T or ARM940T. After arbitrating multiple requests from internal peripherals and GPIO, the Controller requests an interrupt.

The hardware arbitration logic decides the interrupt arbitration process, and the results are recorded in the interrupt pending registers.

7.1.1 Block Diagram



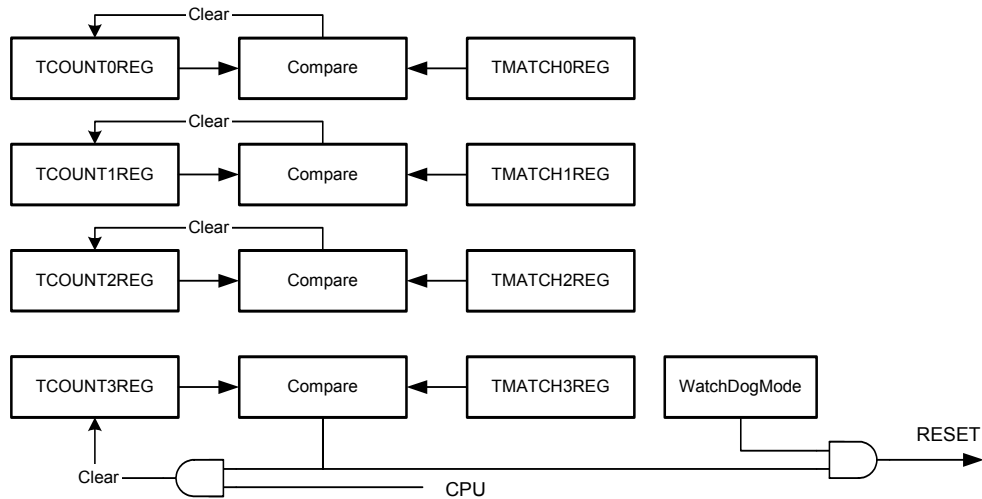
Interrupt Process Diagram

8. Timer

8.1 Overview

There are 4 Timers in VR3D and each Timer operates by divided clock from clock of Clock Generator. Each Timer performs Free running up-counting and is cleared when has same value of each TMATCH*REG(Timer Match Counter).

Timer3 is used as Watchdog Timer. Timer3 Clock should be not divided clock but original clock when it used for Watchdog Timer. Writing 0 in Interrupt Pending Register of Timer3 will clear Watchdog Timer.



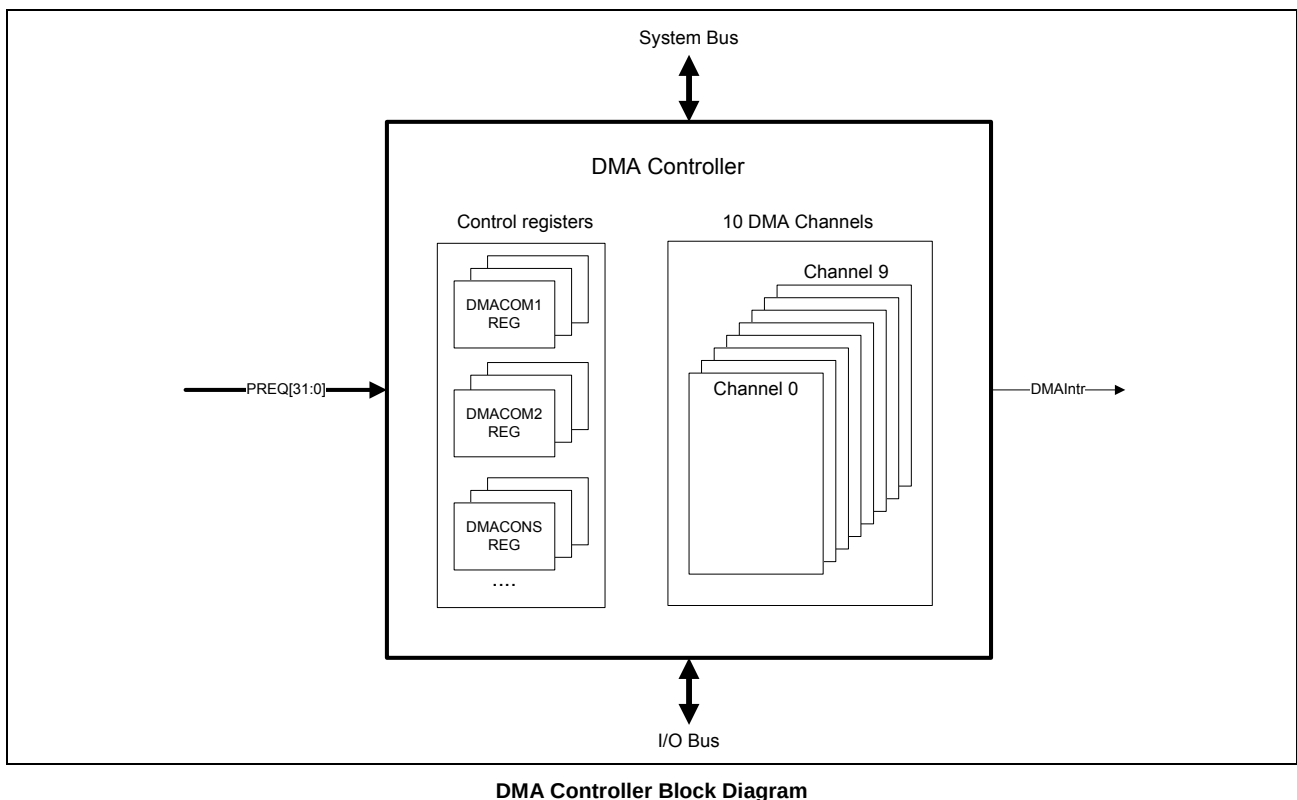
9. DMA

9.1 Overview

The DMA controller has 10 Channels and 4 words or 8 words of burst transmission. In addition, each channel has 8 EA of 32-bit buffers. VR3D supports the 10-channel DMA controller between system buses and peripheral buses. Each channel of the DMA controller can unrestrictedly transmit Data to each device connected to a system bus or a peripheral bus. Therefore, each channel allows devices to transfer data under the following conditions: 1) when the source and the destination are system buses, 2) when the source is a system bus and the destination is a peripheral bus, 3) when the source is a peripheral bus and the destination is a system bus, 4) when the source and the destination are peripheral buses.

The best advantage of the DMA controller is that it can perform the transfer of Data without CPU intervention. The DMA controller performs DMA operation by software, an internal peripheral request or external request.

9.1.1 Block Diagram

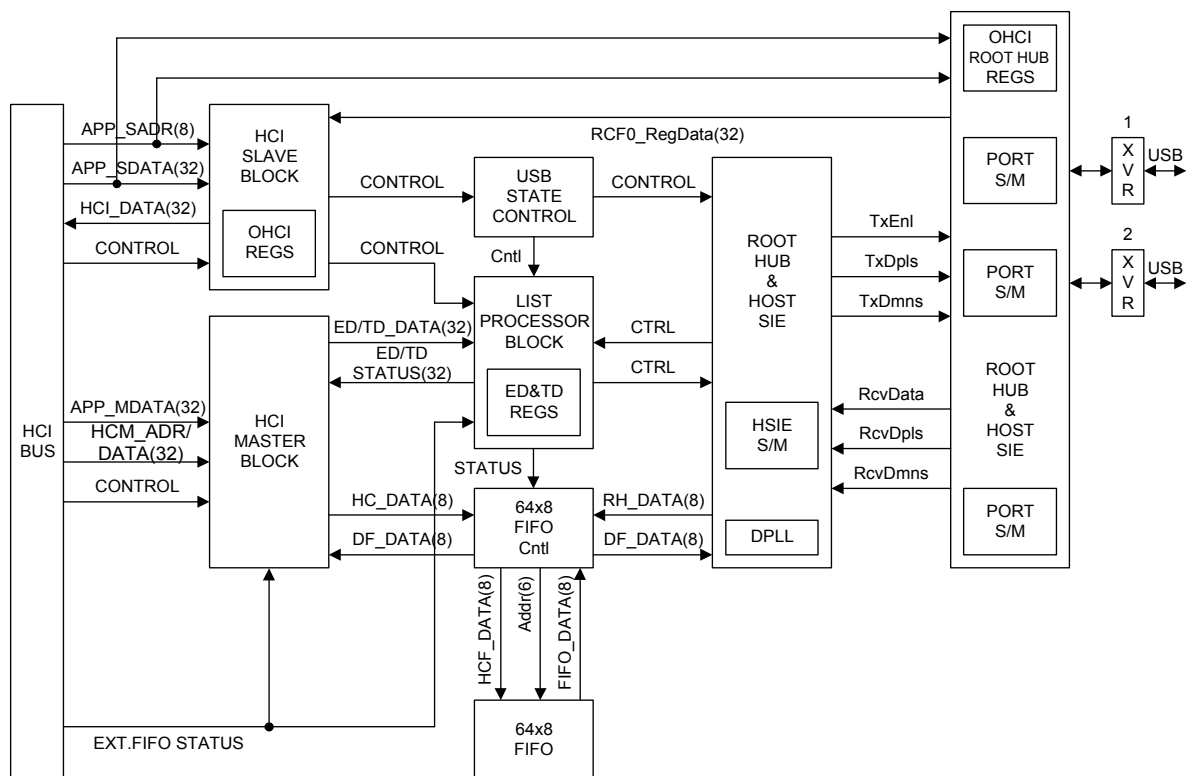


10. USB HOST CONTROLLER (UHC)

10.1 Overview

VR3D supports 2 port USB Host interface as follows

- OpenHCI Rev 1.0 compatible
- USB Rev1.1 compatible
- 2 down stream ports
- Support for both LowSpeed and HighSpeed USB Devices



11. USB DEVICE CONTROLLER (UDC)

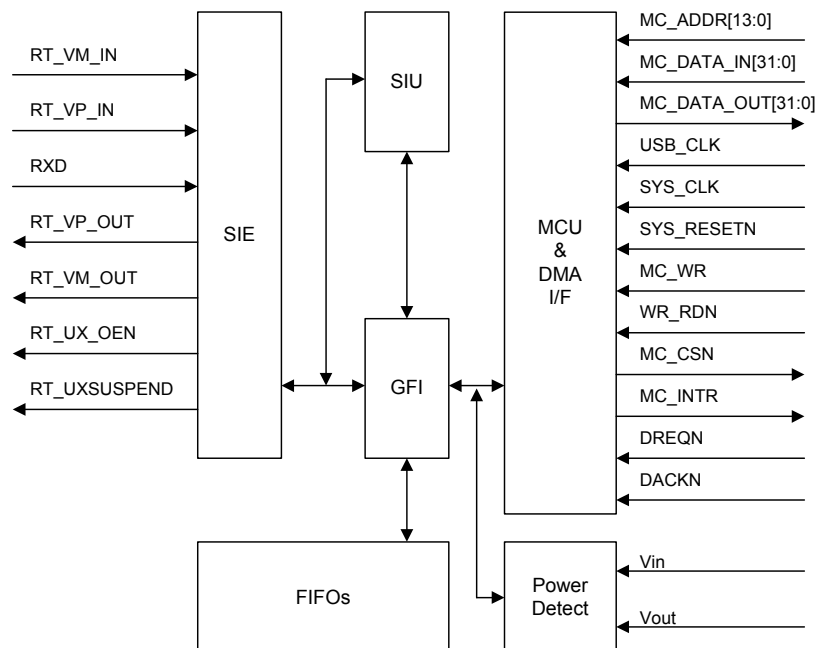
11.1 Overview

USB Function controller is designed to provide a high performance full speed function controller solution with DMA I/F. USB function controller allows bulk transfer with DMA, interrupt transfer and control transfer.

The functions are as follows:

- Full Speed USB Function Controller compatible with the USB Specification Version 1.1
- DMA Interface for Bulk Transfer
- 5 Endpoint with FIFO
 - EP0: 16byte (Register)
 - EP1: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP2: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP3: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP4: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
- Integrated USB Transceiver

11.1.1 Block Diagram



11.1.2 Features

Fully compliant to USB Specification Version 1.1

- Full Speed (12Mbps) Device
- Integrated USB Transceiver
- Supports Control, Interrupt and Bulk transfer
- 5 Endpoint with FIFO:
 - One Bi-directional Control Endpoint with 16 byte FIFO (EP0)
 - Four bi-directional bulk endpoint with 64-byte FIFO (EP1, EP2, EP3, EP4)
- Supports DMA interface for receive and transmit bulk endpoints. (EP1, EP2, EP3, EP4)
- Independent 64byte receive and transmit FIFO to maximize throughput
- Support suspend and remote wake-up function
- USB device operating frequency: 48MHz

12. UART

12.1 Overview

Four independent asynchronous serial I/O (SIO) ports, each of which can operate in interrupt-based or DMA-based mode, are provided by the VR3D UART (Universal Asynchronous Receiver and Transmitter) unit. That is, an interrupt or DMA request to transfer data between the CPU and UART can be generated by the UART. Bit rates of up to 3.1 Mbps can be supported by UART. Two 16-byte FIFOs are contained in each UART channel for receive and transmit.

Programmable baud-rates, one or two stop bit insertion, 5-bit, 6-bit, 7-bit or 8-bit data width and parity checking are included in the VR3D UART.

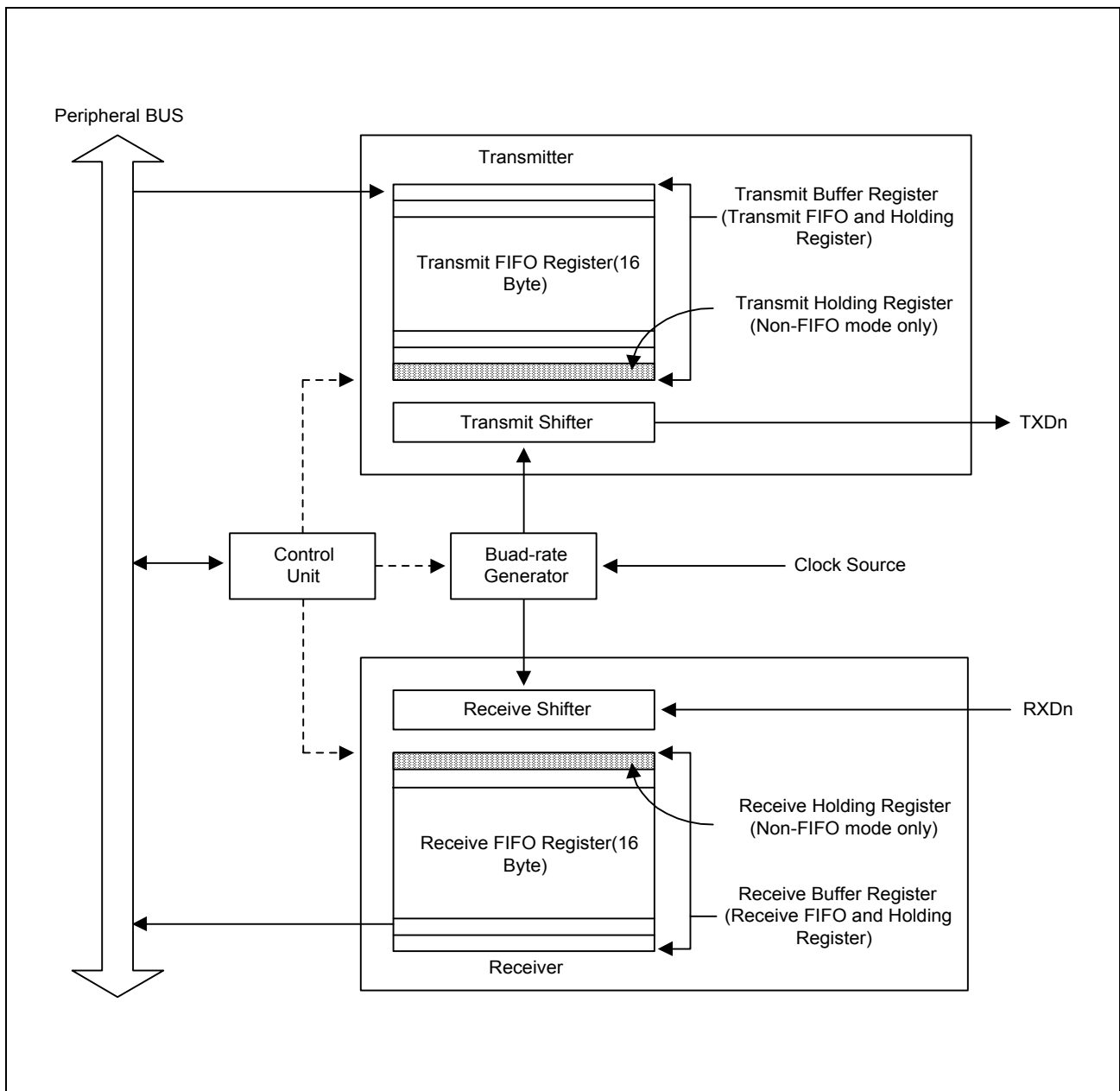
Fig UART Block Diagram (with FIFO) shows that baud-rate generator, transmitter, receiver and control unit are contained in each UART. PCLK can clock the baud-rate generator. The transmitter and the receiver contain 6-byte FIFOs and data shifters. Transmission data is written to FIFO and then copied to the transmit shifter. The transmit data pin (TXDn) then shifts out the data. The received data is shifted from the receive data pin (RXDn), and then copied to FIFO from the shifter.

12.1.1 Features

- RXD0, TXD0, RXD1, TXD1, RXD2, TXD2, RXD3, TXD3 with DMA-based or interrupt-based operation
- UART Ch 1 or Ch2 with full modem function
- UART Ch 2 with IrDA 1.0
- Supports handshake transmit / receive (UART Ch1 or Ch2)
- Ability to add or delete standard asynchronous communications bits (start, stop, and parity) in the serial data
- Independently controlled transmit, receive, line status, and data set interrupts
- Programmable baud rate
- Modem control pins that allow flow control through software.
- Fully programmable serial-interface:
 - 5-, 6-, 7-, or 8-bit characters
 - Even, odd, and no parity detection
 - 1, or 2 stop bit generation
 - Baud rate generation up to around 3.1Mbps
- 16-byte transmit FIFO
- 16-byte receive FIFO
- Complete status reporting capability
- Ability to generate and detect line breaks
- Internal diagnostic capabilities that include:
 - Loopback controls for communications link fault isolation
- Separate DMA requests for transmit and receive data services

Slow infrared asynchronous interface that conforms to the Infrared Data Association (IrDA) standard

12.1.2 Block Diagram



UART Block Diagram (with FIFO)

13. PWM

13.1 Overview

The VR3D has four Pulse Width Modulators (PWM). Each PWM operates respectively and is controlled by its own register set. Each channel outputs the pulse width modulated signal via an external pin. Four PWMs have the same circuit architectures and, hereinafter, will be described as PWMn (n = 0, 1, 2, 3)

Each PWM Channel has a 7-bit Prescaler. The Prescaler receives the Clock provided by the Power Management and divides the Clock with the value specified by the PWMnPERIOD, and then produces the Clock for carrying out the PWM operation.

In addition, each PWM Channel has a Counter composed of 10 bits. The CPU can produce the desired signal by specifying the period and the duty of a PWM Waveform in the PWMnPERIOD and PWMnDUTY registers.

When the specified value in the PWMnPERIOD corresponds to the DUTY Counter value, the PWM resets the Counter and sets the PWM Signal to '0'.

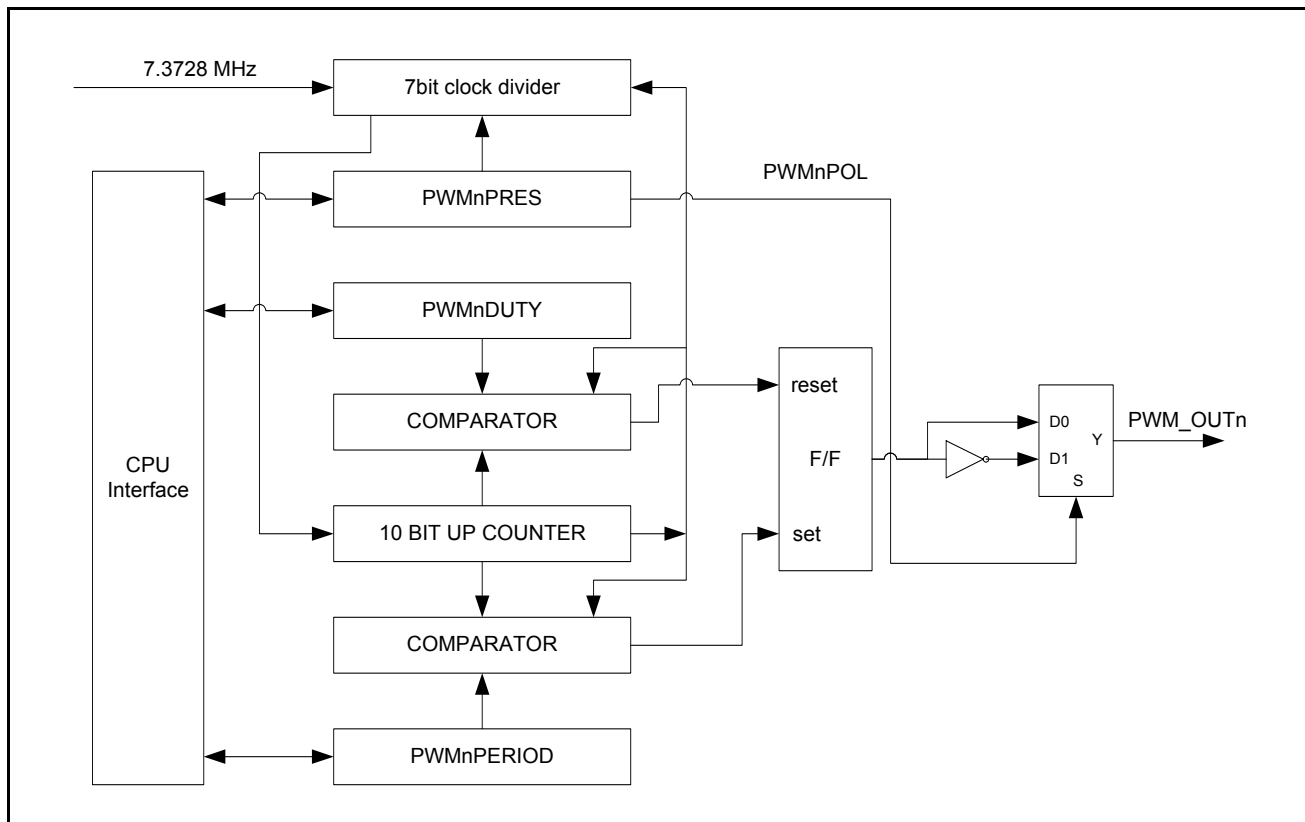
When the specified value in the PWMnDUTY corresponds to the DUTY Counter value, the PWM sets the PWM Signal to '1'.

The PWMnPOL determines whether to Invert or Bypass the value before the PWM Signal produced in the pertinent Channel goes out through the external PAD.

13.1.1 Features

- 4 Pulse Width Modulator channels
- 7-Bit Clock Divider & 10-Bit Period Counter
- 10-Bit Duty Counter

13.1.2 Block Diagram



PWM Block Diagram

14. I2C

14.1 Overview

The MP2520F Microprocessor can support a multi-master I2C-bus serial interface. Information between bus masters and peripheral devices that are connected to the 12C-bus is carried by a dedicated serial data line (SDA) and a serial clock line (SCL). The SDA and SCL lines are bi-directional.

In multi-master I2C-bus mode Serial data to or from slave devices can be received or transmitted by multiple MP2520F Microprocessors. The master MP2520F, by which a data transfer can be initiated over the 12C-bus, is responsible for terminating the transfer. This I2C-bus in MP2520F uses the standard bus arbitration procedure.

Values must be written to the following registers to control multi-master 12C-bus operations:

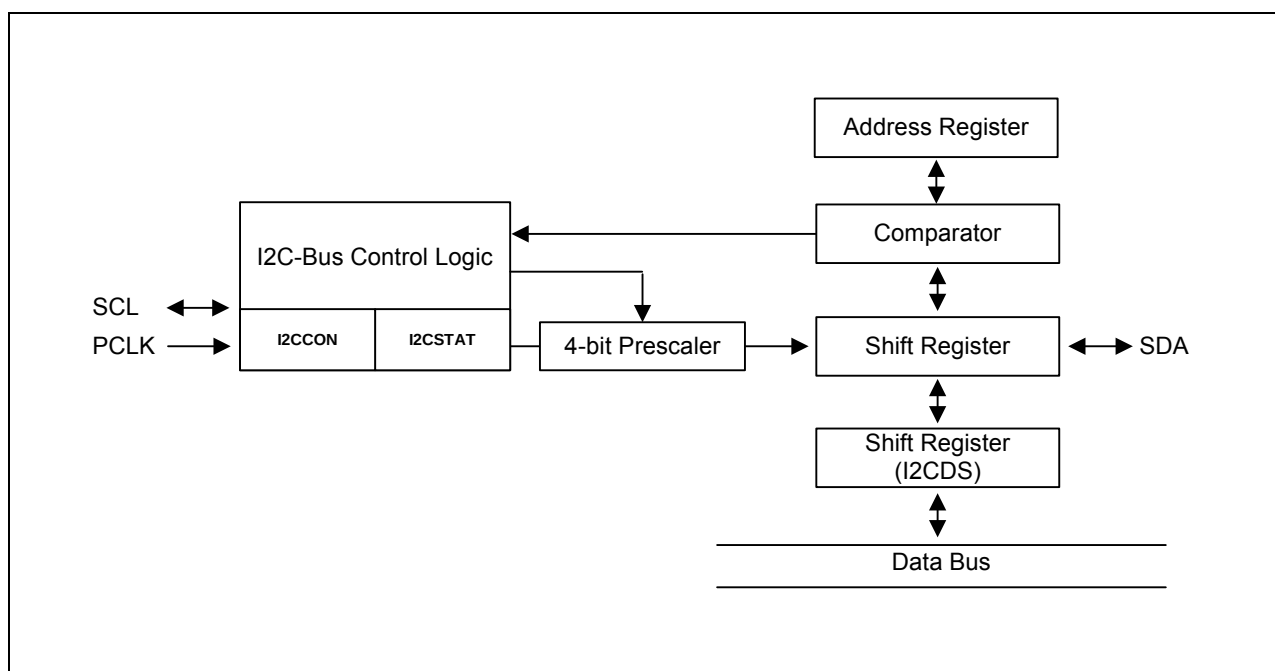
- I2CCON, Multi-master I2C-bus Control Register
- I2CSTAT, Multi-master I2C-bus Control/Status Register
- I2CDS, Multi-master I2C-bus TX/RX Data Shift Register
- I2CADD, Multi-master I2C-bus Address Register

The SDA and SCL lines should both be at High level when the 12C-bus is free. A Start condition can be initiated by a High-to-Low transition of SDA. A Stop condition can be initiated by a Low-to-High transition of SDA while SCL remains steady at High Level.

Start and Stop conditions can always be generated by Master devices. The slave device that the bus master device has selected can be determined by a 7-bit address value in the first data byte, which is put onto the bus after the Start condition has been initiated. The direction of the transfer (read or write) is determined by the 8th bit.

Each SDA line data byte should total eight bits. The number of bytes that can be received or sent during the bus transfer operation is not limited. An acknowledge (ACK) bit should immediately follow every byte, and data is always sent from the most significant bit (MSB) first.

14.1.1 Block Diagram



I2C Block Diagram

15. ADC & TPC

15.1 Overview

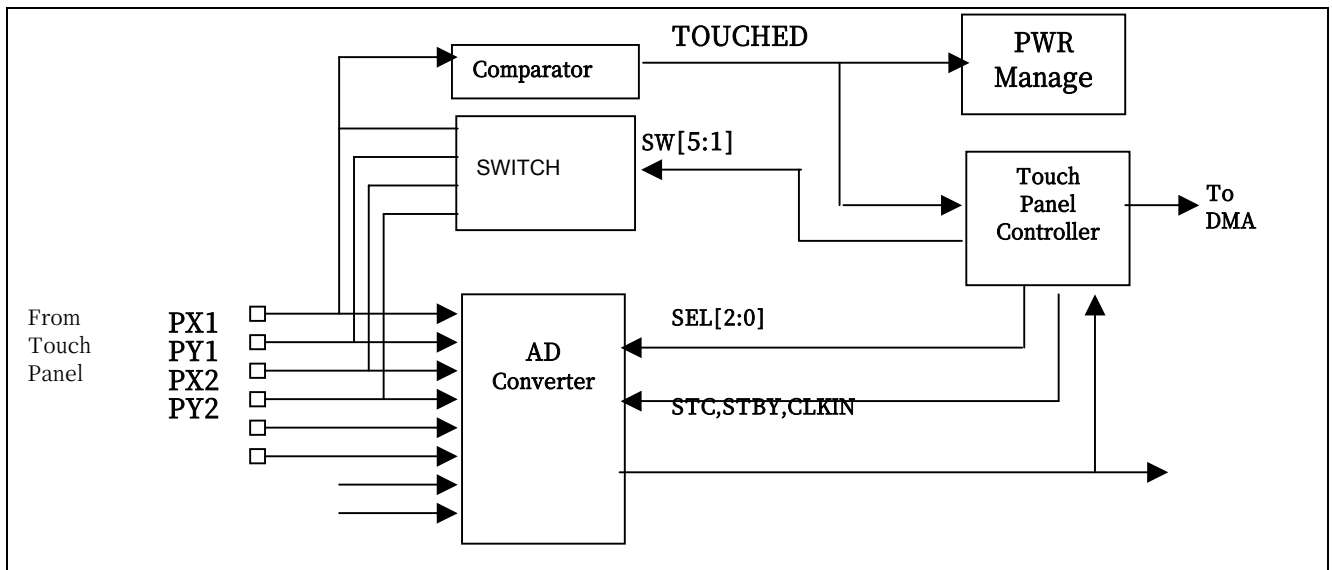
The Touch Panel Controller (TCP) controls the switch, comparator, and ADC to read the X, Y, position of the Touch Panel touched. It also reads the data from any ADC Channel. All data is transmitted into external memory using the DMA. Refer to ADC System Block Diagram.

15.1.1 Features:

The TPC module offers the following features:

- 10-bit ADC for touch panel and voltage detection
- On-chip voltage reference for ADC
- 16x16-bit FIFO for X, Y, Z, U1 and U2
- Embedded touch panel circuitry (Switch and Comparator)
- Supports auto, manual, and Touched Detection Enable Mode (TDEM)
- Programmable pen down and pen-up interrupt to interrupt handler
- Provides data-ready and FIFO-full interrupts

15.1.2 Block Diagram



This block consists of three analog blocks and one control block.

The Comparator detects pen down. The switch creates the bias required for TOUCH PAD when it reads each coordinate. The AD Converter (ADC) converts analog signals entered through a PAD into digital signals. Finally, the TPC (Touch Panel Controller) controls the analog blocks and transfers the data attained from the ADC to the outside through the DMA.

16. SD/MMC Controller

16.1 Overview

Multi-Media Card (MMC), Secure Digital (SD) memory card and SDIO device are supported by the SD/MMC controller. MMC specification version 2.11, SD memory card specification version 1.0 and SDIO card specification version 1.0 are appropriate for the SD/MMC controller.

16.1.1 Feature

- Compatible with the Multi-Media Card System Specification, version 2.11
- Compatible with the SD Memory Card Specification, version 1.0 and SDIO Card Specification, version 1.0
- Supports clock speeds up to 10 MHz in MMC mode
- Supports clock speeds up to 25 MHz in SD mode
- Contains an Internal Clock Prescaler
- Contains 64 bytes of FIFO for data receive/transmit
- Supports PIO and DMA mode data transfer
- Supports 4-bit data bus

16.2 PIN Description

The external SD/MMC card is connected to three (single data line) or six (4 data line) signals.

Name	Direction	Description
SDICLK	O	SD/MMC Interface Clock - Varies due to clock prescaler
SDICMD	B	SD/MMC Command/Response
SDIDAT[3:0]	B	SD/MMC Data

17. NGPIO Controller

17.1 Overview

NGPIO Controller is used for getting the data which external cpu send. NGPIO Data bus (8/16bit) is connected to SRAM bus of external CPU I/F, and NGPIOLAT(or EIRQ3) signal is connected to the chip select signal which is made with address, write enable signal.

When External CPU write data to SRAM bus, The data on SRAM bus is written to FIFO in NGPIO Controller, then The ARM CPU of VR3D can check of validating the data by checking NGPIOSTATE

This Block stores 16bit NGPIO data, sampled at NGPIOLAT signal Rising/Falling Edge, in internal FIFO. It has CPU interface for CPU's reading.

18. SSP/SPI

18.1 Overview

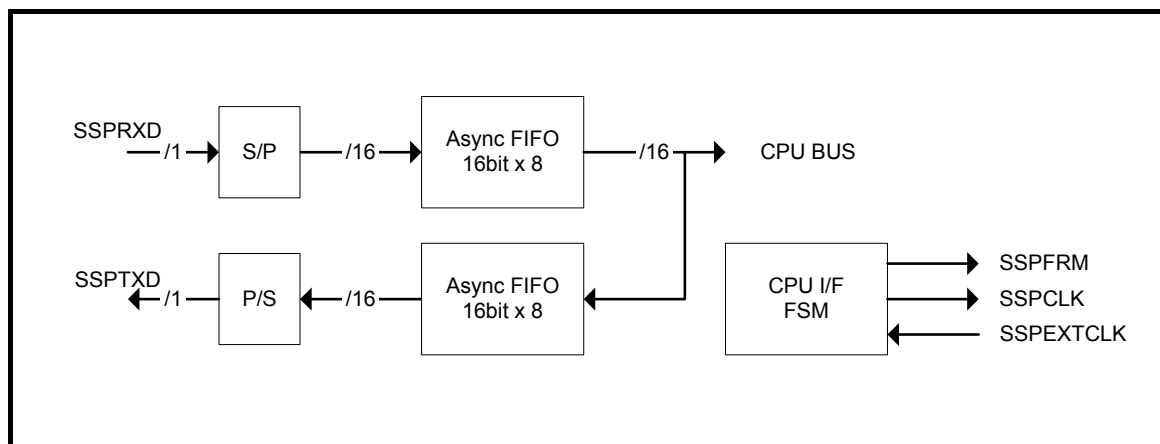
The SSP/SPI is a full-duplex synchronous serial interface. It supports Synchronous Serial Protocol(SSP) and Serial Peripheral Interface(SPI) protocol. It can connect to a variety of external converters, serial memory, and many other devices which use serial protocols for transferring data.

There are 4 I/O pin signals associated with SSP/SPI transfers: the SSPCLK, the SSPRXD data receive line, the SSPTXD data transfer line, and SSPFRM (Frame Indicator in SSP mode, Chip Select in SPI mode).

18.1.1 Features

- SPI Protocol, SSP Protocol compatible
- 16-bit Shift Register for transmit
- 16-bit Shift Register for receive
- 5-bit Pre-scale counter
- Polling, Interrupt, and DMA transfer mode

18.1.2 Block Diagram



SSP/SPI Block Diagram

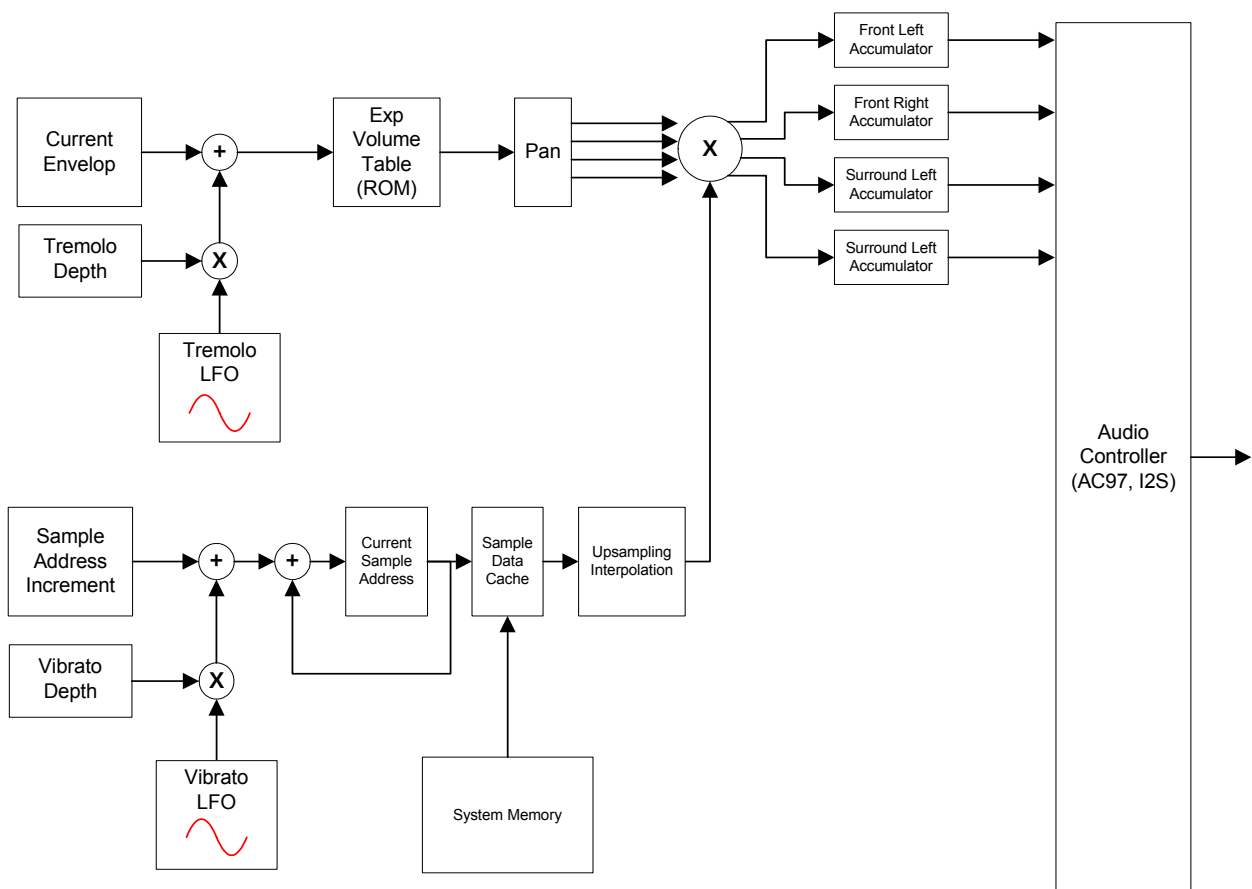
19. Wavetable Synthesizer

19.1 Overview

VR3D has two wavetable synthesizers (WTS). Two WTS are called as WTS#0 and WTS#1. Each WTS can synthesis 32 wave channels. So VR3D can play 64 wave channels. WTS in VR3D has following features

- Two WTS Supports up to 64 wave channels (One WTS supports 32 wave channels)
- Including Vibrato and Tremolo.
- 4 Channel Speaker panning.
- Support speech format (u-LAW) for wave font.

19.1.1 Block Diagram



20. Audio Controller

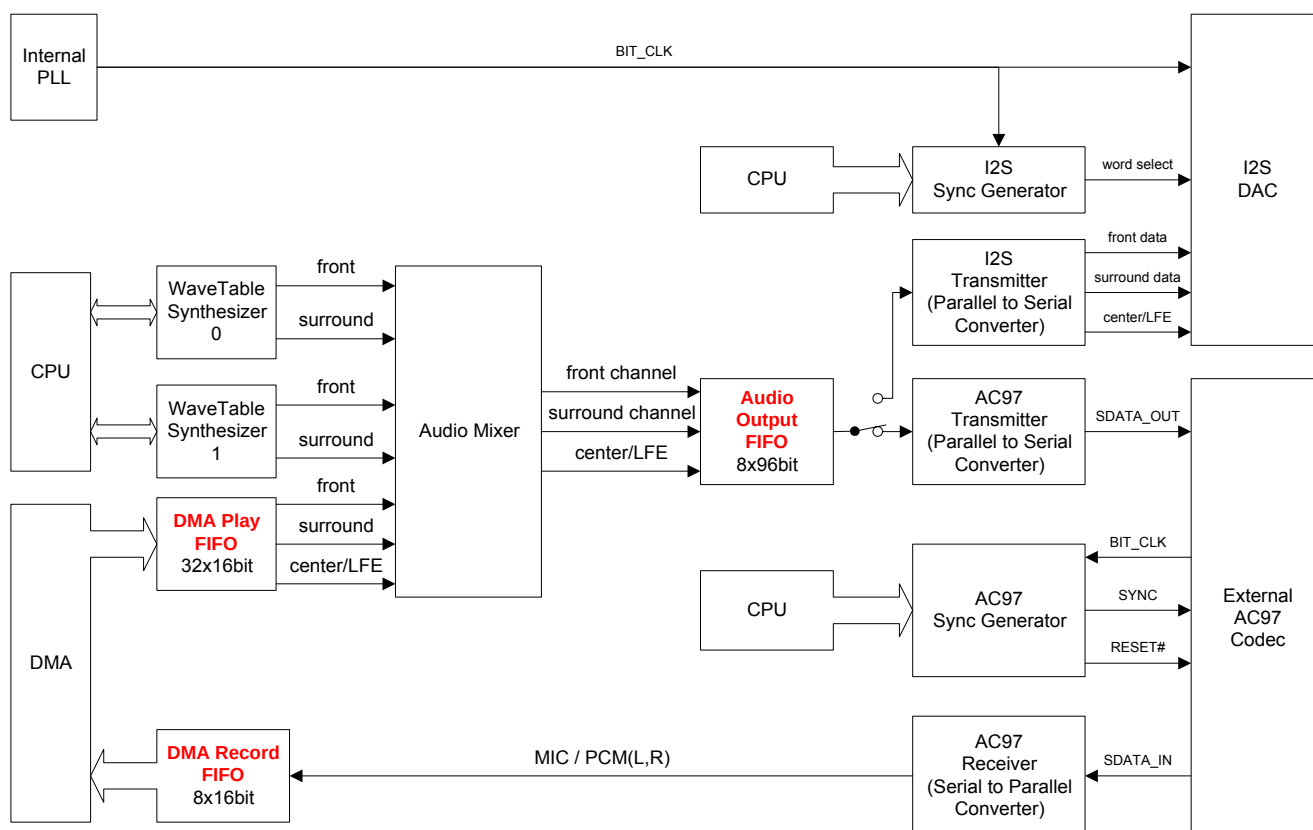
20.1 Overview

Audio Controller controls audio playing, mixing, codec interfacing and recording.

Audio Controller performs followings

- Convert sound data to I2S/AC97
- AC97 codec access
- Audio playing using DMA
- Audio recording using DMA
- Audio playing using wavetable synthesizer (See wavetable synthesizer document)
- Mixing sound data from DMA and wavetable synthesizer

20.1.1 Block Diagram



21. VLD/VLC

The VLD/VLC unit in MMSP-2 receives input video stream(decode) or run/level data(encode) from external memory, and stores decoded run/level data (decode) or video stream (encode) to external memory

21.1 Overview

There are three functional blocks in VLD/VLC unit, SP(Stream Processor), Huffman, and RVLD. SP manipulates in/out bitstream data and are controlled directly by host CPU for header data or by Huffman/RVLD for coefficient data. Huffman block excutes all coefficient VLD/VLC function except for MPEG4 RVLD. RVLD block executes decoding of MPEG4 RVLC code, except for feature reading RVLCs backward. SP is able to be used independently by host CPU for bitstream manipulation for any purpose.

21.1.1 Feature

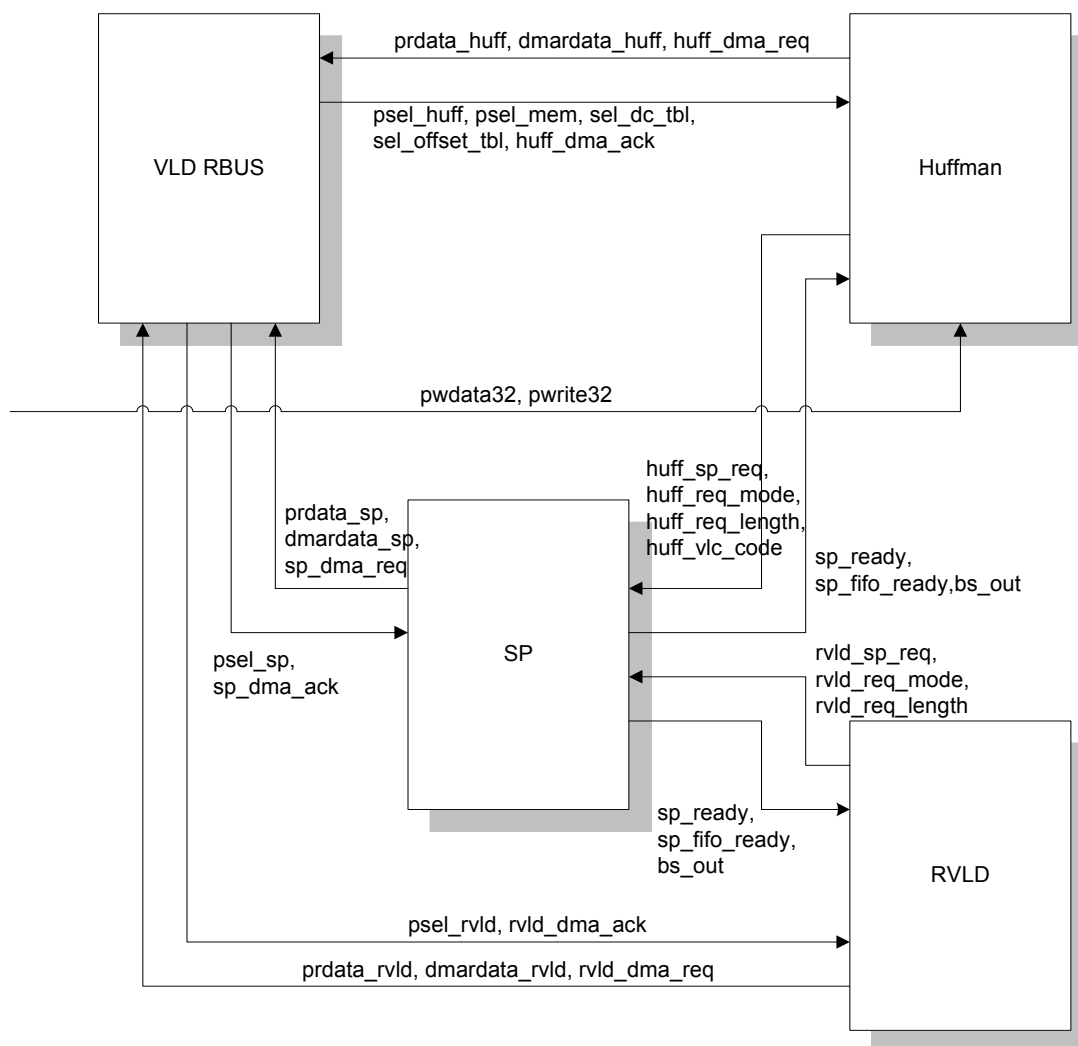
SP(Stream Processor)

- Getbit, Flushbit, Putbit operation with 0 ~ 16 bit length
- Showbit operation with 0 ~ 16 and 32 bit length

VLD/VLC

- MPEG1, MPEG2 and MPEG4 (includes RVLC) decode
- H.263 baseline and Annex I,T decode
- JPEG decode
- MPEG2 and MPEG4 encode
- H.263 baseline encode
- JPEG encode

21.1.2 Block Diagram and Description



- There are two 16word x 32bit FIFOs in SP, VLD/C, RVLD block each used as in/out data buffer.
- SP has 16-bit barrel shifter.
- VLD/VLC has one 32 word x 12 bit, one 32 word x 16 bit, one 64 word x 17 bit register file and 368 word x 21 SRAM for VLC table

22. MPEG4 Picture Decoder

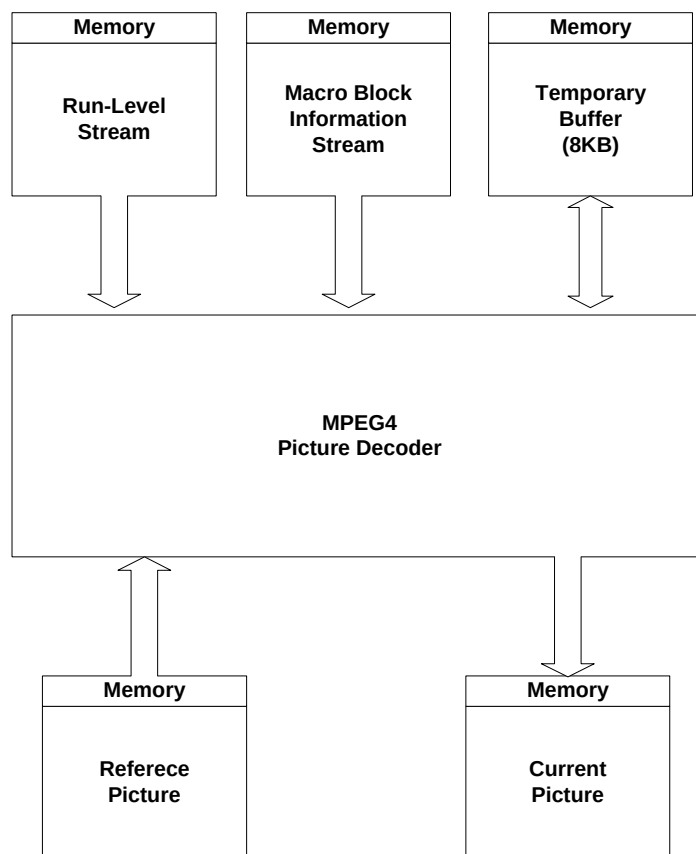
22.1 Overview

MPEG4 decoding process consists of VLD Processor and MPEG4 Picture Decoder. VLD Processor decodes huffman coded bit stream, and MPEG4 Picture Decoder performs following

- Inverse Quantization
- Inverse DCT
- Motion Prediction/Compensation

MPEG4 Picture Decoder requires Run-Level data which is generated from VLD Processor. It is not necessary that you know about VLD Processor because Run-Level data can be generated by programmer (but slow). Please refer to VLD Processor part for more information. This document only explains MPEG4 Picture Decoder. Now we support only MPEG4 Simple Profile. In this document we call MPEG4 Picture Decoder as MPEG4DEC.

22.1.1 Block Diagram



23. 2D Graphic Controller

23.1 Overview

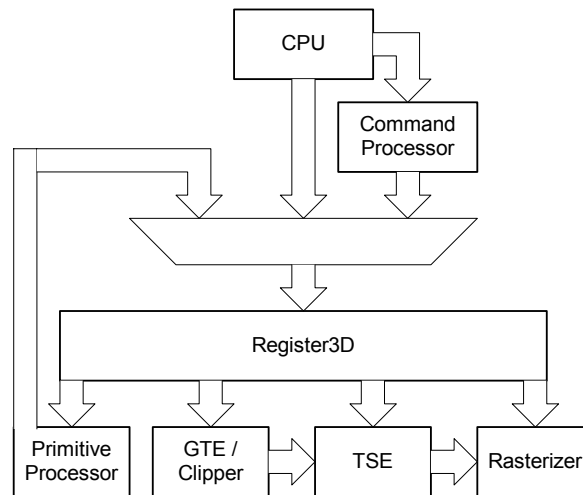
VR3D has 2D Graphic Controller to accelerate GUI and optimized for graphic base O/S such as WindowsCE or Linux

- 32-deep source image FIFO
- Dual-bank Command Register Set provides efficient interface between CPU and graphics controller.
- Hardware support for font color expansion, font caching and image caching
- Graphic acceleration support for 16 bits per pixel mode.
- Graphic acceleration function include :
 - Rectangular source –copy block transfer (BitBLTs)
 - Transparent source-copy BitBLTs
 - Masked source-copy BitBLTs
 - Monochrome-to-color expansion on source and pattern data
 - Pattern and Rectangle Fills
 - 256 Raster Ops (ROPs)

24. 3D Graphic Engine

24.1 Overview

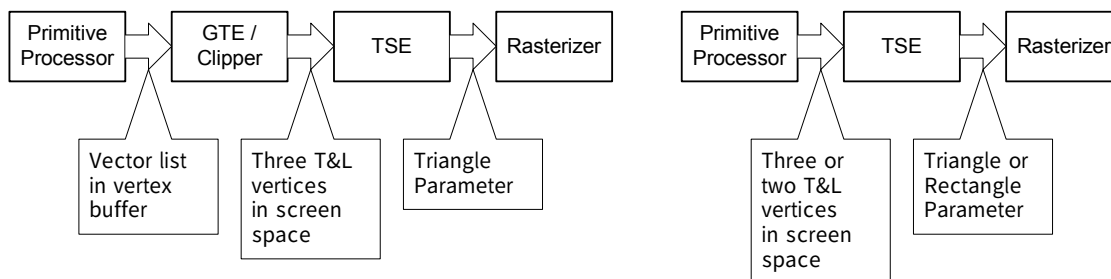
3D core of VRender3D consists of several functional blocks (sub-module).



< 3D Core control flow >

CPU controls all operation of Sub-module or CPU can request Command processor to control them.

Below picture shows data flow of primitive rendering using vertex buffer. Left side is for hardware vertex transform/view volume clipping and right side is for using Transformed & Lit vertex (in screen space).



< 3D Core data flow >

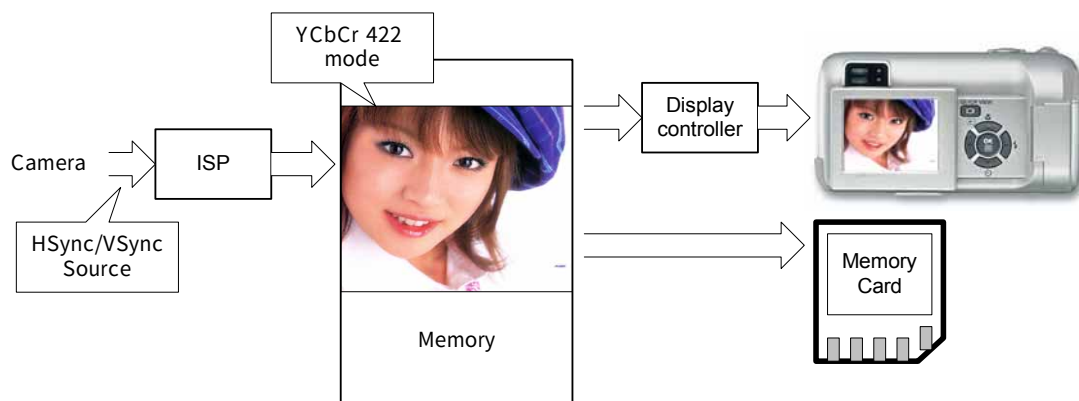
Sub-modules, related primitive rendering, operate with parameters and control signals from Register3D. As special purpose DAM [Command Processor](#), which could write 3D core register, also operate 3D core.

25. Image Signal Processor (ISP)

25.1 Overview

ISP stores YCbCr Image from Camera & External Video to External Memory

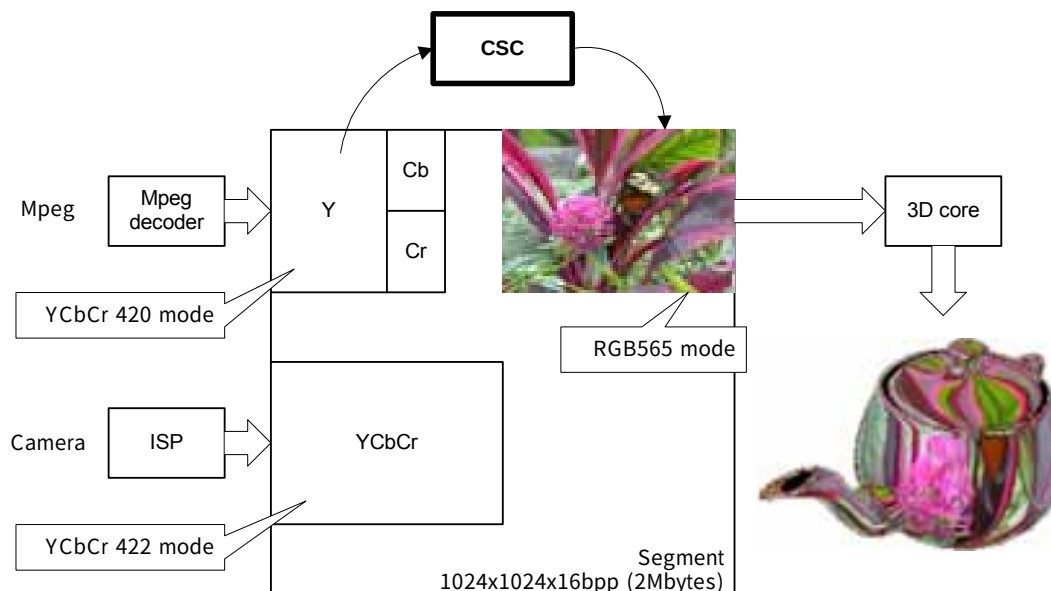
ISP support Linear address and Block address. ISP support course image scale down for preview in real time.



26. Color Space Converter(CSC)

26.1 Overview

Color space converter (CSC) module converts YCbCr output from Mpeg decoder or Camera(ISP) to usable format for using as texture in 3D Core.



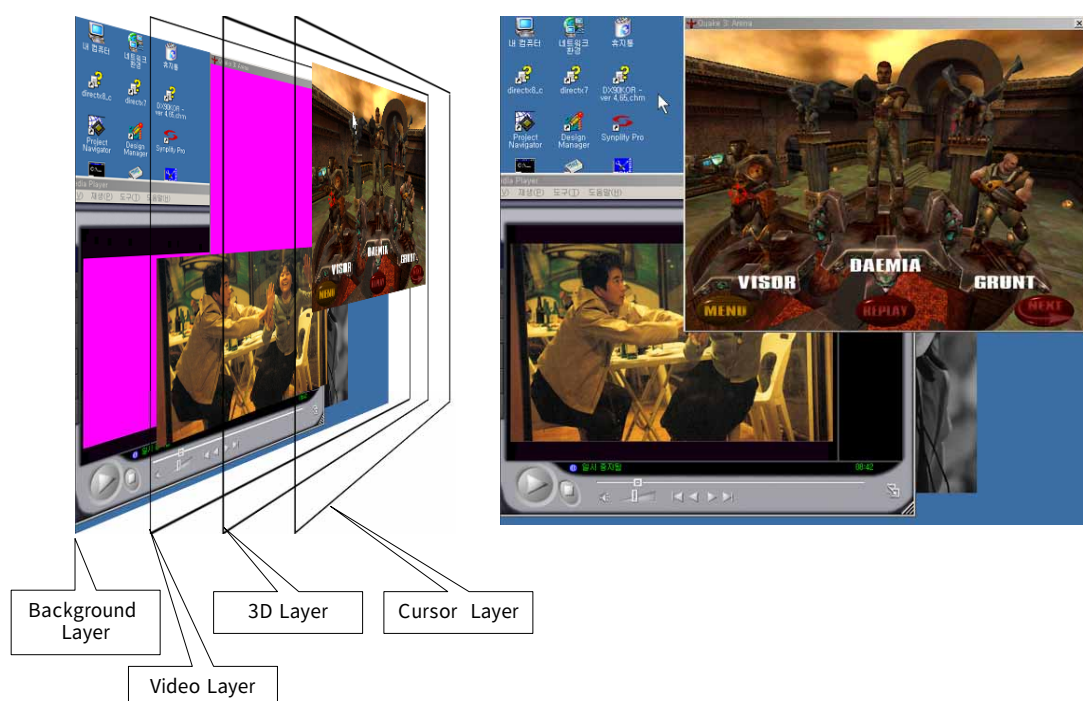
< Color space conversion >

27. Multi Layer Controller (MLC)

27.1 Overview

VR3D can display 4 layers in one Screen at the same time.

Layer name	Pixel format	Addressing mode	Description
Background	RGB565	Linear/Block	Overlay function using ColorKey in 3D/ Video layer
Cursor	RGB565	Linear/Block	Configurable Cursor size Transparency Color / Invert Color
3D	RGB565	Block	Transparency Color Changing buffer address by 3D Core(page flipping)
Video	YCbCr 420 (MPEG) YCbCr 422 (Camera)	Linear/Block	Transparency Color Range(min~max) Scaling(Bilinear filtering)



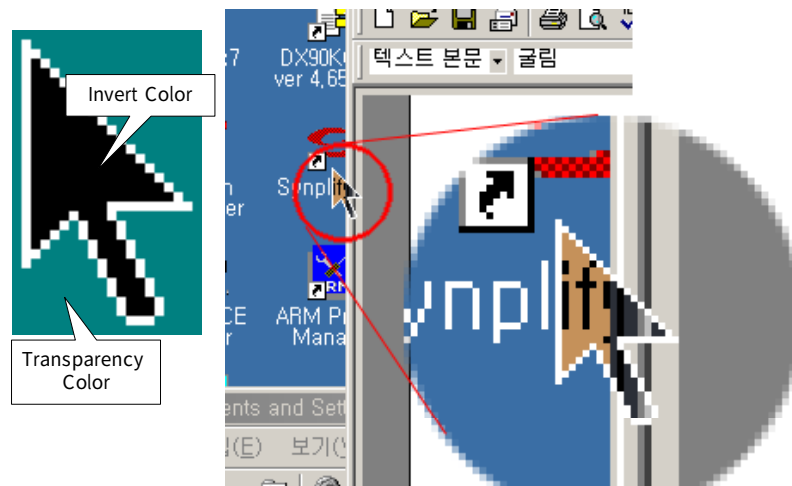
< Display layers >

27.1.1 Background layer

Background layer will be exposed unless other layers cover. Not like other layers, size and position of Background layer are not changeable. (Position is (0,0) and size is same as screen resolution). It can not be located above other layers but it can be exposed above 3D/Video layer when overlay function is used. When overlay is active, only part of 3D/Video layer, filled with certain color, will be exposed to screen.

27.1.2 Cursor layer

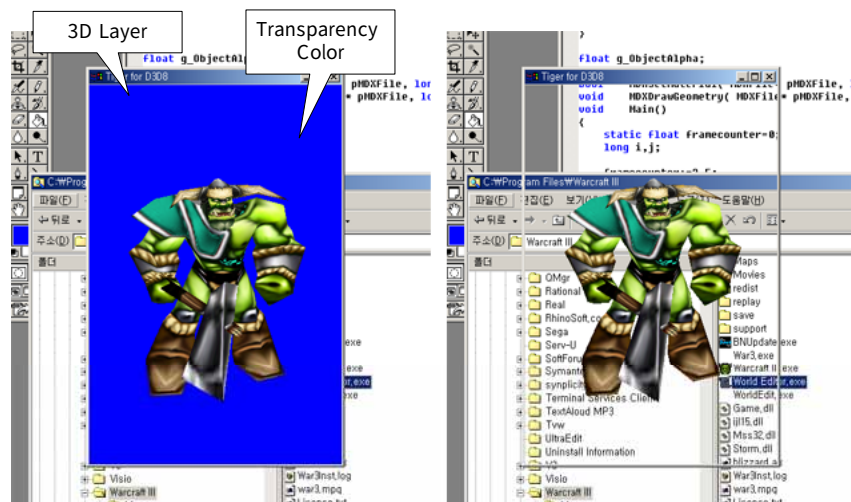
Cursor layer locates top of all layer. The size and position can be changeable. It also set to transparency and opposite color.



< Transparency color & Invert color >

27.1.3 3D layer

3D layer is for 3D frame buffer display. The size and position can be changed. Frame buffer context can not be scaled. Transparency option exist. All or part of 3D layer can be blocked by background layer if overlay function is active.

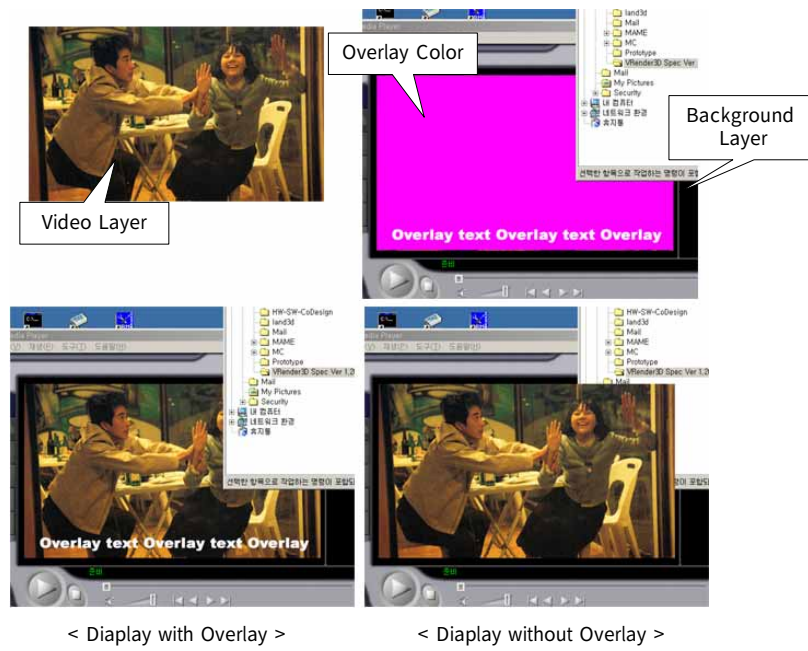


< 3D layer without Transparency >

< 3D layer with Transparency >

27.1.4 Video layer

Video layer display YCbCr pixel format. The size and position can be changeable. The contents can be scaled to fit window size. Transparency option exist. All or part of Video layer can be blocked by background layer if overlay function is active. Supported pixel formats are YCbCr422 mode and YCbCr420 mode. (refer to [Appendix](#))



Note : VR3D can not support filtered YUV layer in interlaced mode.

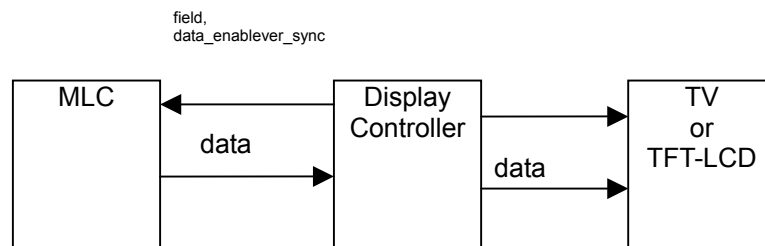
28. Display Controller

28.1 Overview

The Display Controller (DPC) creates signals and data to connect to external display devices. Such external devices are TFT LCD, Video Encoder, DAC, etc. The connection to external devices requires various kinds of Synchronized Signals and Data Formats. The DPC can be set to a variety of Modes according to the types of external display devices. In addition, this block creates signals to be used for the Display FIFO in the Multi Layer Controller (MLC)

Feature

- Support for Display size of up to 1024x768 pixels
- Programmable HSYNC/VSYNC generation
- Supports Composite Sync
- Programmable polarity for DE, HSYNC, VSYNC
- Programmable Pixel Clock Phase
- Supports RGB, Multiplexed RGB(MRGB)



Display Controller

29. ELECTRICAL DATA

29.1 ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Rating		Unit
DC Supply Voltage	VDDi	1.8V Vdd	2.7	V
	VDDIO	3.3V Vdd	3.8	
DC Input Voltage	Vin	3.3V Input Buffer	3.8	
		3.3V Interface / 5V Tolerant input buffer	6.5	
DC output Voltage	VOOUT	3.3V Output buffer	3.8	
DC Input (Latch-up) Current	Iin	± 200		mA
Storage Temperature	Tstg	-65 to 150		°C

29.2 RECOMMENDED OPERATION CONDITIONS

Parameter	Symbol	Rating		Unit
DC Supply Voltage	VDDi	1.8V Vdd	1.65~1.95	V
		3.3V Vdd	3.0~3.6	
DC Supply Voltage for I/O Block	VDDi	1.8V Vdd	1.65~1.95	
		3.3V Vdd	3.0~3.6	
		3.3V Vdd	3.3 $\pm$ 5%	
DC Input Voltage	VIN	3.3V Input buffer	3.3 $\pm$ 0.3	
		3.3V Interface / 5V Tolerant input buffer	3.8 ~5.25	
DC Output Voltage	VOOUT	3.3V Output buffer	3.3 $\pm$ 0.3	
Operating Temperature	TOPR	Commercial	0 to 70	°C

29.3 D.C. ELECTRICAL CHARACTERISTICS

CHARACTERISTIC

These DC Parameters guarantee the I/O cell characteristic at the static state only, not at the dynamic state.

29.3.1 Normal I/O – 1.8v

Vdd = 1.65 ~ 1.9v, Ta = 25°C

Characteristics	Symbol	Test Condition		Min	Typ	max	unit
Input High Current	Iih	Vin = Vdd	Normal	-10	-	10	uA
Input Low Current	Iil	Vin = Vss	Normal	-10	-	10	uA
Input High Voltage	Vih	CMOS		1.27	-	-	V
Input Low Voltage	Vil	CMOS		-	-	0.57	V
Output High Voltage	Voh	0mA Buffer, Ioh= - 0mA		1.2	-	-	V
Output High Voltage	Voh	3mA Buffer, Ioh= - 3mA		1.2	-	-	V
Output High Voltage	Voh	20mA Buffer, Ioh= - 20mA		1.2	-	-	V
Output Low Voltage	Vol	0mA Buffer, Ioh= 0mA		-	-	0.45	V
Output Low Voltage	Vol	3mA Buffer, Ioh= 3mA		-	-	0.45	V
Output Low Voltage	Vol	20mA Buffer, Ioh= 20mA		-	-	0.45	V

Normal 1.8v I/O PAD DC Electrical Characteristics

29.3.2 Normal I/O – 3.3v

Vdd = 3.0 ~ 3.6v, Ta = 25°C

Characteristics	Symbol	Test Condition		Min	Typ	max	unit
Input High Current	Iih	Vin = Vdd	Normal	-10	-	10	uA
Input Low Current	Iil	Vin = Vss Pull-up	Normal Up	-10 -60	-	10 - 10	uA
Input High Voltage	Vih Vt+	CMOS Schmitt S (CMOS)		2.0 -	- -	- 2.0	V
Input Low Voltage	Vil Vt-	CMOS Schmitt S (CMOS)		- 0.8	- -	0.8 -	V
Output High Voltage	Voh	8mA Buffer, Ioh= - 8mA		2.4	-	-	V
Output High Voltage	Voh	12mA Buffer, Ioh= - 12mA		2.4	-	-	V
Output High Voltage	Voh	16mA Buffer, Ioh= - 16mA		2.4	-	-	V
Output Low Voltage	Vol	8mA Buffer, Ioh= 8mA		-	-	0.4	V
Output Low Voltage	Vol	12mA Buffer, Ioh= 12mA		-	-	0.4	V
Output Low Voltage	Vol	16mA Buffer, Ioh= 16mA		-	-	0.4	V
Tri-State Output Leakage Current	Ioz	Vout = Vdd or Vss		-10	-	10	uA

Note : Schmitt Trigger test condition : Vdd = 3.0 to 3.6 v, Ta = 25°C

Normal 1.8v I/O PAD DC Electrical Characteristics

29.3.3 DDR Normal I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Input High Voltage	V _{ih}	CMOS	1.43	-	-	V
Input Low Voltage	V _{il}	CMOS	-	-	1.07	V
Output High Voltage	V _{oh}	26mA Buffer, I _{oh} = - 26mA	2.138	-	-	V
Output Low Voltage	V _{ol}	26mA Buffer, I _{ol} = 26mA	-	-	0.114	V

Comment : psbstl, psosstl, V_{ref}=1.25V

DDR Normal I/O PAD DC Electrical Characteristics

29.3.4 DDR Differential I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Output High Voltage	V _{oh}	26mA Buffer, I _{oh} = -26mA	2.138	-	-	V
Output Low Voltage	V _{ol}	26mA Buffer, I _{ol} = 26mA	-	-	0.114	V

Comment : psosstldiff, V_{ref}=1.25V

DDR Differential I/O PAD DC Electrical Characteristics

29.3.5 Special I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Input High Voltage	V _{t+}	SCHITT (CMOS)	2.0	-	-	V
Input Low Voltage	V _{t-}	SCHITT (CMOS)	-	-	0.8	V
Output High Voltage	V _{oh}	8mA Buffer, I _{oh} = - 8mA	2.4	-	-	V
Output Low Voltage	V _{ol}	8mA Buffer, I _{ol} = 8mA	-	-	0.4	V

Comment : phbsu100ct8sm, phbsu100ct12sm, Controllable Pull-Up

Special I/O PAD DC Electrical Characteristics

29.3.6 USB I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	Unit
Input High Voltage	V _{ih}		2.5	-	-	V
Input Low Voltage	V _{il}		-	-	0.8	V
Input High Current	I _h	V _{in} = 3.3V	-10		10	uA
Input Low Current	I _l	V _{in} = 0.0V	-10		10	uA
Output High Voltage	V _{oh}	15KΩ to GND	2.8	-	3.6	V
Output Low Voltage	V _{ol}	1.5KΩ to 3.6v	-	-	0.3	V

USB PAD DC Electrical Characteristics

29.3.7 Normal Operation

Parameter	Value	Unit	Condition
Operation Voltage	1.8/2.5/3.3	V	
Max. Operating Frequency (FCLK)	200	MHz	CPU Clock
Max. Operating Frequency (BCLK)	100	MHz	Bus Clock
Max. Operating Frequency (PCLK)	50	MHz	Internal Peripheral Clock
Typical normal mode power (VDDi Only)	390	mW	Full 3D Engine Operation ARM 200Mhz MPEG Core Operation
Idle mode power (VDDi Only)	160	mW	CPU Stop, Other Logic running.
Stop mode power (VDDi Only)	0.5	mW	All Clock Stop

VRender3D Power Supply Voltage and Current